

Trustworthy Federated Graph Neural Networks: Unifying Fairness, Privacy, Robustness, and Explainability for High-Stakes Risk Detection

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Abstract

Federated Learning (FL) enables institutions to train shared models over sensitive, decentralised graph data—transactions, credit records, social ties—without exchanging raw data. Yet accuracy alone does not make such systems deployable: recent regulatory and scientific frameworks (the EU AI Act, the NIST AI Risk Management Framework) demand that automated decision systems be simultaneously *fair*, *private*, *robust*, and *transparent*. Existing work addresses these properties in isolation, and no prior method combines fairness, differential privacy, and Byzantine robustness for federated graph neural networks (GNNs). We present **TrustFedGNN**, a unified trustworthy federated graph-learning framework built from three components: (i) *Fairness-Sensitive Edge Reweighting* (FSER), which suppresses the biased cross-group message passing responsible for structural discrimination; (ii) *Fairness-Targeted Gradient Decomposition* (FTGD), which applies differential privacy to the *released fairness statistic* by privatising only the low-dimensional demographic-parity means—so strong DP for the fairness signal costs $\mathcal{O}(\mathbf{1})$ noise where full-gradient DP-SGD must inject $\mathcal{O}(\sqrt{|\theta|})$ and destroys utility (we state precisely what this guarantee covers, and what it does not, in the text); and (iii) *Bi-objective Frank–Wolfe Aggregation* (BFWA) and its Byzantine-robust variant, which impose an operator-chosen constraint on the aggregated client-disparity statistic while screening malicious

clients, including a new fairness-poisoning threat. We further instrument the framework with predictive uncertainty, GNN attention audits, a composite trust score, sustainability accounting, and an EU AI Act / NIST RMF compliance mapping. Across four demographic benchmarks (German, Credit, Bail, Pokec-z) and two large-scale scalability studies (Elliptic Bitcoin, 203,769 nodes; ogbn-products, 2.4M nodes), TrustFedGNN does not dominate every individual metric, but *to our knowledge*, it is the only method among sixteen baselines to combine competitive fairness and utility with a statistic-level sensitive-attribute DP guarantee and Byzantine robustness in a single federated GNN under simulated cross-silo partitions. All tables and figures are regenerated from logged runs and the code is released, so every reported number is traceable to a configuration. This work supports SDG 9 by advancing trustworthy collaborative AI.

Keywords: Federated learning, Graph neural networks, Algorithmic fairness, Differential privacy, Byzantine robustness, Explainable AI, Trust metrics, Regulatory compliance

1 Introduction

Consider a consortium of banks that together want to catch money laundering. Fraud is fundamentally *relational*: illicit funds move through chains of accounts, shell companies, and intermediaries, and it is precisely the *structure* of these money flows—not any single transaction in isolation—that betrays criminal intent. Graph Neural Networks (GNNs) have therefore become the tool of choice for financial crime detection, credit scoring, and other high-stakes risk assessments, because their message-passing mechanism learns directly from the web of relationships between users, accounts, and transactions [1–3]. Yet the very banks that would benefit most from a shared model are legally forbidden from pooling their customers’ data: privacy regulations such as the GDPR, and plain commercial confidentiality, keep each institution’s transaction graph locked inside its own walls. Federated Learning (FL) offers a way out. By exchanging model updates instead of raw data, a group of institutions can train a single, more powerful detector without any of them surrendering its records [4].

This picture, however, is dangerously incomplete. A model that is merely *accurate* is no longer fit to be deployed in decisions that shape people’s access to credit, liberty, or financial services. Three converging forces—the European Commission’s Ethics Guidelines for Trustworthy AI, the EU AI Act, and the U.S. NIST AI Risk Management Framework—now demand that automated decision systems be—depending on their risk classification, the provider’s role, and the jurisdiction—*fair, privacy-preserving, robust, transparent, and accountable*. For systems that fall under the EU AI Act’s high-risk category, several of these move from good practice toward regulatory obligation (subject to the Act’s scope and conformity-assessment provisions, which we do not attempt to adjudicate here). Meeting them is hard even for a centralised model. In the federated, graph-structured setting they collide, and each one becomes markedly harder than in isolation.

Why the federated graph setting is uniquely hard.

The difficulty is not that we must bolt four independent features onto a classifier; it is that the natural solution to each requirement actively sabotages the others. *Fairness* is confounded by the graph itself: homophilous message passing means that a member of a demographic group who happens to be connected to risky neighbours will be flagged as risky too, so the model learns to launder a protected attribute through graph structure—and the central server, which never sees any data, cannot even measure this bias, let alone correct it. *Privacy* appears to have a textbook answer, differentially private SGD, but that answer fails badly here: adding calibrated noise to every coordinate of the gradient injects perturbation whose magnitude grows with the square root of the model dimension, and on the compact GNNs used for tabular financial graphs this noise simply drowns the signal, collapsing accuracy to little better than chance (we quantify this in Section 5.6). *Robustness* is threatened not only by the classical malfunctioning or malicious client, but—as we show—by a subtler adversary that the fairness machinery itself invites: a client can transmit a deliberately biased model while *reporting* exemplary fairness statistics, thereby hijacking any aggregator that rewards fairness. And *transparency* is undermined by the opacity of relational models, exactly when regulation requires that each adverse decision be explainable and auditable.

The gap.

The research literature has, understandably, attacked these problems one at a time. Fair-GNN methods deliver impressive demographic parity but assume all data sits in one place [5–7]. Fair federated learning has matured for tabular and image data but rarely touches graphs [8, 9]. Byzantine-robust aggregation hardens FL against poisoning but is oblivious to fairness [10, 11]. Differentially private FL protects data but, as noted, at a utility cost that is prohibitive for graph models. To the best of our knowledge, *no* existing method delivers fairness, differential privacy, and Byzantine robustness together for federated GNNs, and none has been validated on a large-scale cryptocurrency graph—arguably the most consequential modern fraud domain. Closing this gap is the goal of this paper.

Our approach.

We present **TrustFedGNN**, a single framework in which fairness, privacy, robustness, and transparency are designed to reinforce rather than undermine one another. Its architecture (Fig. 1) intervenes at the three natural control points of federated graph training—inside message passing, at the local gradient, and at server aggregation—and wraps the result in an instrumentation layer that measures, explains, and audits trustworthiness. The key technical insight, and the reason privacy becomes almost free, is a change of perspective on *where* the sensitive attribute actually influences learning: it does so only through a two-dimensional summary statistic, so we can protect that statistic directly instead of blindly noising the entire gradient.

Concretely, our contributions are:

1. **Privacy-Aware Fairness Feedback Loop (C1).** We introduce a closed-loop co-design coupling Fairness-Targeted Gradient Decomposition (FTGD, Section 4.2)

with Bi-objective Frank–Wolfe Aggregation (BFWA, Section 4.3). FTGD establishes that sensitive attributes influence the global fairness objective solely through low-dimensional demographic-parity means, enabling rigorous (ϵ, δ) -differential privacy over the released fairness statistics at $O(1)$ noise without the catastrophic $O(\sqrt{|\theta|})$ utility collapse of full-gradient DP-SGD. When differential privacy is active (`dp_mode='ftgd'`), each client releases the noised disparity statistic $\widetilde{\text{DPD}}_k = |\tilde{\mu}_0 - \tilde{\mu}_1|$ actually produced by its local FTGD training step, and BFWA and the other metadata-consuming aggregators operate on this privatised signal; without DP, clients report the (un-noised) validation-split DPD. BFWA’s penalised Frank–Wolfe program then steers global aggregation *toward* an operator-chosen disparity budget τ on these reported client statistics, with per-round feasibility reported as a residual rather than guaranteed (Section 4.3).

2. **Byzantine-Robust Fairness Aggregation (C2).** We uncover and defend against a deceptive fairness-poisoning threat model where adversarial clients fabricate zero-disparity statistics while transmitting biased or corrupted model updates (Section 5.8). To defend against this, we develop `robust_bfwa`, pairing coordinate-wise distance screening with constrained dual reweighting. At the client level, Fairness-Sensitive Edge Reweighting (FSER, Section 4.1) provides structural regularization by learning attention corrections that suppress biased cross-group edges, preventing topological bias amplification before local updates are computed.
3. **Multi-Dimensional Trustworthiness Evaluation & Empirical Grounding (C3).** We conduct an extensive empirical audit across four demographic benchmarks (German, Credit, Bail, Pokec-z) and two large-scale scalability studies (Elliptic Bitcoin, 203,769 transactions; ogbn-products, 2.4M nodes) against sixteen baselines (Section 5). Our evaluation provides rigorous ablation grids, multi-seed Byzantine robustness sweeps, an adaptive stealth adversary breakdown analysis, non-IID Dirichlet and topological community scalability tests, 2,000-configuration Monte Carlo trust score rank stability, centralized sanity anchors, and non-demographic operational subgroup audits, fully integrated with EU AI Act and NIST RMF compliance artifacts (Section 6).

Relation to a preliminary version.

A short preliminary version of this work, introducing the FSER, FTGD, and BFWA components under the name FedFairGNN, appeared as a conference paper [12] evaluated on three fraud-detection graphs (YelpChi, Amazon, Elliptic) with a single $K=3$ -client configuration and no robustness, uncertainty, explainability, or regulatory-compliance analysis. This article substantially extends that preliminary study: it (i) evaluates on five standard fairness benchmarks (German, Credit, Bail, Pokec, Elliptic) plus a 2.4M-node scalability study (ogbn-products), only one of which overlaps with the earlier evaluation; (ii) compares against sixteen baselines, including five 2025–2026 methods reimplemented for this comparison, versus the small baseline set of the preliminary version; (iii) introduces the Byzantine-robust `robust_bfwa` variant and a new fairness-poisoning attack, entirely absent from the earlier work; (iv) adds an adversarial evaluation of the privacy mechanism (an attribute-inference attack,

Section 5.6) and multi-seed statistical significance testing, neither of which the preliminary version performed; and (v) adds the full trustworthiness instrumentation layer—predictive uncertainty and calibration, attention-based audits, a composite trust score, sustainability accounting, and the EU AI Act / NIST AI RMF design mapping (Section 6)—which did not appear in the preliminary version at all. We also renamed the framework from FedFairGNN to TrustFedGNN to reflect this broadened scope.

By lowering the barrier to trustworthy, infrastructure-grade collaborative AI, this work contributes to UN Sustainable Development Goal 9 (industry, innovation, and resilient infrastructure). The remainder of the paper is organised as follows. Section 2 situates our work within the fairness, privacy, and robustness literatures; Section 3 formalises the federated graph setting and the trustworthiness objectives; Section 4 develops the TrustFedGNN framework component by component; Section 5 reports the empirical study; Section 6 analyses trustworthiness holistically and maps the system to regulation; and Section 7 discusses implications, limitations, and broader impact before we conclude.

2 Related Work

Fairness in graph neural networks.

GNNs inherit and amplify bias through homophilous message passing. Remedies include adversarial debiasing (FairGNN [5]), counterfactual and stability regularisation (NIFTY [13]), pre-processing that minimises inter-group distributional distance (EDITs [14]), sensitive-information neutralisation via heterogeneous neighbours (FairSIN [6]), re-balancing through counterfactual mixup (FairGB [15]), invariant learning across inferred sensitive partitions (FairINV [16]), and disentangled debiasing (DAB-GNN [7]). All assume *centralised* data. Our FSER is closest in spirit to the edge-reweighting variants of FairSIN and to distribution-aware reweighting [17], but operates inside a federated, private, and robust pipeline.

Fairness in federated learning.

Two notions coexist. *Client (performance) fairness* equalises accuracy across clients: q-FedAvg [9] up-weights high-loss clients. *Group fairness* equalises outcomes across demographic groups: FairFed [8] adjusts aggregation weights by each client’s local fairness gap; FedFB ports FairBatch to FL; PUFFLE [18] blends a demographic-parity loss with DP-SGD under a target-disparity controller; FedFACT [19] gives provable global-and-local group fairness via a Lagrangian saddle point over a shared global dual variable and per-client local duals; Bendoukha et al. [20] (PoPETs’25) make FairFed’s weighting homomorphically computable, replacing its $\exp(-\beta|\cdot|)$ term with an FHE-friendly degree-2 polynomial and aggregating under threshold CKKS; and EquFL [21] (2026) adds a server-side fairness-gradient correction on top of an arbitrary aggregation rule, provably compatible with Byzantine-robust ones. These target tabular or image data. For *graphs*, F²GNN [22] combines a fairness-weighted aggregation with structure-aware group-balance weights. The most recent efforts sit closest to our setting: FairGFL [23] adds DP to federated fair GNNs but targets *cross-client*

(performance) fairness arising from imbalanced overlapping subgraphs rather than demographic group fairness; FedGraph-Fair [24] applies distributionally-robust optimisation for personalised group-fair federated graph learning; FaVGNN [25] (2026) achieves group fairness in *vertical* FL via completion-driven adversarial fusion; and FDP-Fair [26] (2026) combines federated DP training with demographic-parity post-processing. A preliminary version of this work by two of the present authors [12] introduced the FSER, FTGD, and BFWA components under the name FedFairGNN on a smaller evaluation without a robustness, uncertainty, or compliance analysis; this article is a substantial extension, detailed in Section 1. We compare against FairFed, q-FedAvg, FedFB, F²GNN, FairSIN (run federated), the strongest 2026 competitors (horizontal adaptations of FaVGNN and FDP-Fair), and—reimplementing each paper’s core statistical mechanism to the extent it transfers to a single-graph, star-topology federation (Appendix B)—FairGFL, FedGraph-Fair, PUFFLE, FedFACT, and the PoPETs’25 FairFed variant, none of which had a public head-to-head against a fairness-aware federated *GNN* prior to this comparison.

Differential privacy in FL.

DP-FedAvg [27] clips per-client updates and adds Gaussian noise; privacy is tracked with the moments accountant / Rényi DP [28, 29]. On the compact GNNs used for tabular graphs, isotropic gradient noise at meaningful ϵ overwhelms the signal (Section 5.6). FedFDP [30] adapts clipping to fairness but still privatises the whole gradient, and CE-FedGNN [31] (2026) instead targets communication efficiency under a metric-DP guarantee for federated GNN training—neither touches the fairness statistic specifically. FTGD instead privatises only the *sufficient statistics* of the fairness term, exploiting the fact that the sensitive attribute never enters the task loss—a targeted mechanism whose (ϵ, δ) guarantee follows from the Gaussian mechanism and post-processing immunity. A further 2026 result is cautionary: locally private GNNs remain vulnerable to structure-inference adversaries even under a formal ϵ budget [32], which is why we couple FTGD’s privacy guarantee with the robust aggregator (BFWA/robust-BFWA) rather than treating DP alone as a sufficient defence.

Byzantine-robust aggregation.

Krum and Multi-Krum [10], coordinate-wise median and trimmed mean [11], Bulyan [33], and trust-bootstrapping schemes such as FLTrust [34] defend against malicious updates under attacks including sign-flipping, model-replacement [35], ALIE [36], and IPM [37]. None targets fairness. Recently, fairness-poisoning attacks emerged—EAB-FL [38] and the fairness-constrained optimisation attack [39]—that *increase* disparity while preserving accuracy, evading both robust and fairness-aware defences. We reproduce the fairness-constrained optimisation attack of [39] and defend with a robust fairness-constrained aggregator. Two further 2026 advances harden aggregation against stronger threat models without touching fairness: CQSA [40] clusters updates before a quantum-secure aggregation step, and a post-quantum lattice-based scheme [41] combines adaptive weighting with CRYSTALS-Kyber encryption for critical-infrastructure deployments; both are, like their predecessors, fairness-blind.

Explainability, uncertainty, and trust metrics.

GNN explainers attribute predictions to edges/features; we use the FSER attention itself as a native, inspectable audit signal (not a validated prediction-level explanation) and add gradient-based attribution of the *disparity*. Predictive uncertainty via MC-dropout [42] and calibration (ECE, Brier) quantify confidence reliability; a concurrent 2026 approach, FedWQ-CP [43], instead calibrates conformal-prediction sets in a single round under dual (statistical and system) heterogeneity—a complementary, distribution-free alternative to our MC-dropout-based calibration that we discuss as future work (Section 7). Composite trust scoring for FL is nascent; we contribute a transparent power-mean aggregation over utility, fairness, privacy, robustness, and calibration. On the compliance side, CryptoFair-FL [44] (2026) uses homomorphic encryption to verify demographic parity and equalised odds without exposing sensitive attributes, explicitly mapping to EU AI Act Articles 10 and 13—independent confirmation that regulatory-grounded, cryptographically verifiable fairness auditing is an active 2026 research direction, though it does not integrate Byzantine robustness or an explainability layer as we do. Finally, we align the system to the EU AI Act and NIST AI RMF, operationalising accountability as an auditable artifact rather than prose.

Positioning.

Table 1 places the most relevant methods on the five axes that define our problem. The pattern is stark: the fair-GNN methods are strong on graph fairness but centralised; the fair-FL methods federate but ignore graph structure; robust aggregation hardens FL but is fairness-blind; and even the newest federated fair-graph methods (F²GNN, FairGFL, FaVGNN, FedGraph-Fair) each leave at least one axis uncovered—typically Byzantine robustness, which no fairness-aware federated method addresses. Among the methods surveyed here, TrustFedGNN is the only entry that combines all five axes at once—federated, graph-native, group-fair, differentially private (for the fairness statistic), and Byzantine-robust—and the only one we evaluate on a large-scale cryptocurrency graph (using, as noted, a temporal proxy rather than a demographic attribute). It also reports the fullest trustworthiness picture—uncertainty, attention audits, sustainability, and a regulatory design-mapping—within a single framework. This axis-coverage claim is about *breadth of integration*, not about being best on any single axis, where, as Section 5.3 shows, specialised baselines can win individual cells.

3 Preliminaries and Problem Setting

Cross-silo federated graph learning.

There are K institutions (clients). Client k holds a private graph $\mathcal{G}_k = (\mathcal{V}_k, \mathcal{E}_k, \mathbf{X}_k, \mathbf{y}_k, \mathbf{s}_k)$ with node features $\mathbf{X}_k \in \mathbb{R}^{n_k \times d}$, binary task labels $\mathbf{y}_k \in \{0, 1\}^{n_k}$ (e.g. fraud/default/recidivism), and a binary *sensitive attribute* $\mathbf{s}_k \in \{0, 1\}^{n_k}$ (e.g. gender, age, race). Raw graphs never leave the client. A server coordinates T communication rounds; each round clients perform E local epochs and transmit model parameters, which the server aggregates. We evaluate the global model on the pooled held-out test nodes of all clients. Statistical heterogeneity is modelled by a symmetric

Table 1: Where methods sit across the trust axes. ✓: supported; ~: partial/adaptable; blank: not addressed. “Graph” = graph-native; “Group fair” = demographic group fairness (not merely client fairness); “DP (stat.)” = differential privacy; “Crypto” = validated on a large-scale cryptocurrency graph. [†]Guaranteed strictly over released fairness statistics, not full model updates (Section 4.2).

Method	Fed.	Graph	Group fair	DP (stat.)	Byz.	Explain	Crypto
FairGNN [5]		✓	✓			~	
NIFTY [13]		✓	✓				
FairSIN [6]		✓	✓				
DAB-GNN [7]		✓	✓				
FairFed [8]	✓		✓				
q-FedAvg [9]	✓		~				
PUFFLE [18]	✓		✓	✓			
FedFACT [19]	✓		✓				
ByITFL [45]	✓		✓		✓		
DP-FedAvg [27]	✓			✓			
Krum/Median [10, 11]	✓				✓		
F ² GNN [22]	✓	✓	✓				
FairGFL [23]	✓	✓	~	✓			
FaVGNN [25]	✓	✓	✓	✓			
FDP-Fair [26]	✓		✓	✓			
FedGraph-Fair [24]	✓	✓	✓				
PoPETs’25 [20]	✓		✓	✓			
TrustFedGNN (ours)	✓	✓	✓	✓ [†]	✓	✓	✓

Dirichlet(α) split over labels, the standard FL non-IID model [46]; smaller α is more skewed.

Group fairness.

For predicted probabilities $\hat{y}$ and sensitive attribute s we report three group-fairness metrics (lower is fairer):

$$\text{DPD} = |\mathbb{E}[\hat{y} \mid s=0] - \mathbb{E}[\hat{y} \mid s=1]| \quad (\text{Demographic Parity Difference}), \quad (1)$$

$$\text{EOD} = |\text{TPR}_{s=0} - \text{TPR}_{s=1}| \quad (\text{Equal Opportunity Difference}), \quad (2)$$

$$\Delta_{\text{EOdds}} = \max(|\Delta \text{TPR}|, |\Delta \text{FPR}|) \quad (\text{Equalized Odds}). \quad (3)$$

DPD is computed on soft scores (matching the differentiable training term); EOD and Equalized Odds use thresholded predictions.

Differential privacy.

A randomised mechanism $\mathcal{M}$ is (ϵ, δ) -differentially private if for all adjacent datasets D, D' and all outputs S , $\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \Pr[\mathcal{M}(D') \in S] + \delta$ [47]. We protect

the *sensitive attribute*: adjacency is defined by flipping one node’s s_i . The Gaussian mechanism adds $\mathcal{N}(0, \sigma^2)$ noise calibrated to the L_2 sensitivity of the released quantity; composition across rounds is tracked with Rényi DP [28]. For the Gaussian mechanism with noise multiplier $z = \sigma/\Delta$, the Rényi divergence of order α is $\epsilon_{\text{RDP}}(\alpha) = \alpha/(2z^2)$; over T releases it composes as $T\alpha/(2z^2)$, and the tight conversion to (ϵ, δ) -DP is $\epsilon = \min_{\alpha>1} [\epsilon_{\text{RDP}}(\alpha) + \log(1/\delta)/(\alpha - 1)]$.

Threat model for robustness.

Up to $f < K/2$ clients are Byzantine and may send arbitrary updates. We consider data poisoning (label flipping), classical model poisoning (Gaussian, sign-flip, scaling/model-replacement [35]), the strong omniscient attacks ALIE [36] and IPM [37], and a *fairness-poisoning* attack: a malicious client transmits a biased, scaled update while reporting fabricated fairness/utility statistics ($\text{DPD} \approx 0$, high AUC) to capture a fairness-aware aggregator. The server is honest but curious and never accesses raw data.

Objective.

We seek a global model that jointly (i) maximises task utility (AUC-ROC), (ii) minimises group unfairness, (iii) satisfies a target (ϵ, δ) -DP for the sensitive attribute, and (iv) retains utility and fairness under Byzantine corruption—the trustworthiness trade-off that motivates Section 4.

4 The TrustFedGNN Framework

The design philosophy behind TrustFedGNN is that trustworthiness should be engineered where each threat originates, not retrofitted afterwards. Structural bias is born inside message passing, so we correct it there; the sensitive attribute leaks through the gradient, so we privatise exactly the part of the gradient that carries it; bias and poisoning are amplified at aggregation, so we constrain and screen the aggregation step. The framework therefore places three lightweight interventions at the three natural control points of federated graph training and wraps them in a measurement layer that renders the result auditable. Figure 1 gives the end-to-end picture: in each round the server broadcasts the global parameters θ_{t-1} ; every client runs E local epochs of an FSER-augmented GNN and, through FTGD, releases its fairness statistic under differential privacy (reducing, though not eliminating, the update’s sensitive-attribute footprint—Section 4.2); the server then screens the updates for Byzantine behaviour and aggregates the survivors with BFWA under an operator-chosen fairness target to produce θ_t . We now develop each component, motivating it before formalising it.

4.1 Fairness-Sensitive Edge Reweighting (FSER)

Intuition.

Message passing is the source of a GNN’s power and, for fairness, of its peril. Each layer updates a node by mixing in its neighbours’ representations, so a node inherits the statistical fate of the company it keeps. When the graph is homophilous along

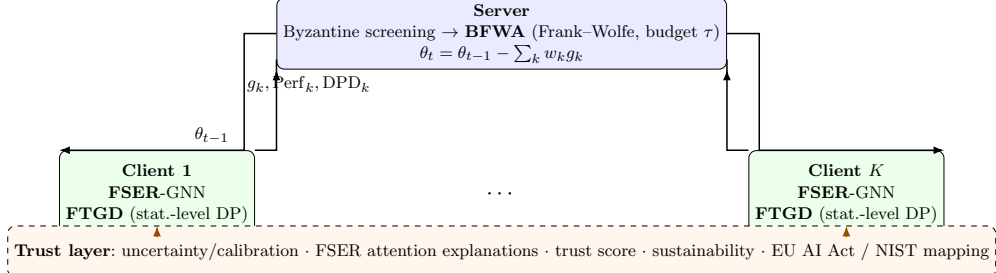


Fig. 1: TrustFedGNN architecture. Three interventions act at complementary control points—FSER inside client message passing, FTGD at the local gradient, and BFWA (with Byzantine screening) at server aggregation—while a trust layer instruments the whole system for audit. Only model weights are transmitted, so fairness and privacy add no communication overhead.

a protected attribute—members of one group disproportionately connected to one another, or to flagged neighbours—the network can encode group membership into embeddings, and the downstream classifier can exploit it without ever being told the sensitive attribute. Bias can propagate through both same-group and cross-group edges; FSER targets one specific, tractable pattern—a *cross-group* edge whose endpoints have become highly similar in representation space. Our hypothesis is that such edges disproportionately carry *demographic* rather than behavioural similarity, so down-weighting them reduces disparity at low utility cost. We do not claim to causally separate demographic from behavioural similarity—high cosine similarity can arise from genuinely shared behaviour, class, or neighbourhood—so FSER is best read as a *learnable heuristic* whose value is established empirically (Section 5.4), not as a certified causal debiaser. It is also, deliberately, only one of three interventions: FTGD and BFWA act on the pathways FSER leaves untouched.

Mechanism.

FSER adds a learnable correction to graph attention. For edge (i, j) with head embeddings $\mathbf{h}_i, \mathbf{h}_j$, the standard attention logit $e_{ij} = \text{LeakyReLU}(\mathbf{a}^\top [\mathbf{h}_i \| \mathbf{h}_j])$ is corrected by a *fairness risk score*

$$\phi_{ij} = \mathbb{I}(s_i \neq s_j) \cdot \max(0, \cos(\mathbf{h}_i, \mathbf{h}_j)), \quad \tilde{e}_{ij} = e_{ij} - \beta^{(l)} \phi_{ij}, \quad (4)$$

so that *cross-group* edges between similarly-embedded nodes are down-weighted after the per-node softmax. The suppression strength $\beta^{(l)} \geq 0$ is learned per layer (clamped to $[0, 5]$). Note that FSER evaluates $\mathbb{I}(s_i \neq s_j)$ and therefore requires the protected attribute of both edge endpoints at training *and* inference time: s is excluded from the node feature matrix $\mathbf{X}$ (so the classifier is not fed s as an input feature) but is used by the attention mechanism. This is a real deployment requirement—and a governance nuance we make explicit in Section 6—rather than a case of the model being blind to s . Because ϕ_{ij} and the resulting attention are retained, FSER also exposes an *inspectable*

audit signal: Section 6 reports how much attention mass FSER removes from cross-group edges. We are careful not to call this a faithful explanation—attention weights are a diagnostic of the mechanism’s behaviour, not a validated attribution of individual predictions (see the explainability discussion in Section 6).

4.2 Fairness-Targeted Gradient Decomposition (FTGD)

The DP-SGD dilemma.

Differentially private SGD is the standard way to keep a client’s data from leaking through its updates, and in principle it is exactly what we need. In practice it is a poor fit here. DP-SGD clips the entire gradient to a norm bound C and adds isotropic Gaussian noise to every one of the $|\theta|$ coordinates; the total noise injected therefore grows like $\sigma\sqrt{|\theta|}$. For the large over-parameterised networks of vision and language this is tolerable, because the signal is spread across many redundant directions. The compact GNNs used on tabular financial graphs are the opposite regime: a few tens of thousands of parameters, every one of them carrying signal. At any privacy level strong enough to be meaningful, the noise required simply overwhelms the gradient, and accuracy collapses toward chance (Section 5.6 quantifies this). Blindly privatising the whole gradient is thus self-defeating. The question FTGD asks is: *do we actually need to?*

Objective.

The local objective is $\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda \mathcal{L}_{\text{fair}}$, with weighted BCE task loss and the differentiable soft demographic-parity penalty $\mathcal{L}_{\text{fair}} = |\mu_0 - \mu_1|$, where $\mu_g = \frac{1}{n_g} \sum_{i:s_i=g} \hat{y}_i$ is the group- g mean prediction.

Key observation.

The sensitive attribute s enters the fairness objective only through the two scalar group means (μ_0, μ_1) : the task loss is a function of $(\mathbf{X}, \mathbf{y})$ alone, and s ’s contribution to $\mathcal{L}_{\text{fair}}$ is fully summarised by these two bounded-sensitivity statistics. FTGD therefore privatises *the released fairness statistic* with a Gaussian mechanism, so that strong DP for the fairness signal costs $O(1)$ noise rather than the $O(\sqrt{|\theta|})$ noise a full-gradient mechanism would inject. We are careful, below, to state precisely which released object this guarantee covers—and which sensitive-attribute pathways it does *not*.

Mechanism.

We treat the group counts (n_0, n_1) as public (they are needed to calibrate the noise scale, and each client’s split enforces a public minimum group size $n_{\min}$). Under the adjacency that reassigns one node’s group label with the counts held fixed, and using $\hat{y}_i \in [0, 1]$, each mean changes by at most $1/n_g$, so (μ_0, μ_1) has L_2 sensitivity $\Delta = \sqrt{1/n_0^2 + 1/n_1^2} \leq \sqrt{2}/n_{\min}$. We release $\tilde{\mu}_g = \mu_g + \mathcal{N}(0, (z\Delta)^2)$ with noise multiplier z , form $\mathcal{L}_{\text{fair}} = |\tilde{\mu}_0 - \tilde{\mu}_1|$, and backpropagate. To reduce task/fairness gradient conflict we orthogonalise the task gradient against the fairness gradient (gradient surgery) before the step.

Proposition 1 (Privacy of the released fairness statistics) *Assume: (A1) model predictions are bounded in $[0, 1]$ via a sigmoid output layer; (A2) local group counts (n_0, n_1) are publicly fixed with $n_{\min} = \min(n_0, n_1) \geq 1$; (A3) the adjacency relation $\mathcal{D} \sim \mathcal{D}'$ reassigns the sensitive attribute s_i of at most one node while preserving total counts; and (A4) the global L_2 sensitivity is bounded by $\Delta = \sqrt{1/n_0^2 + 1/n_1^2} \leq \sqrt{2}/n_{\min}$. Then the per-step release of $(\tilde{\mu}_0, \tilde{\mu}_1)$ satisfies the Gaussian mechanism at sensitivity Δ with noise scale $\sigma = z\Delta$, achieving $(\alpha, \alpha/(2z^2))$ -RDP with respect to s . Over T releases across R rounds and E local epochs ($T = RE$), the composed mechanism satisfies $(\alpha, T\alpha/(2z^2))$ -RDP, converting to:*

$$\left(\min_{\alpha > 1} \left[\frac{T\alpha}{2z^2} + \frac{\log(1/\delta) + \log \alpha}{\alpha - 1} + \log \left(1 - \frac{1}{\alpha} \right) \right], \delta \right) \text{-DP} \quad (5)$$

for the sequence of released fairness statistics.

Proof sketch By (A1) and (A3), altering the sensitive attribute of one node shifts at most one prediction $\hat{y}_i \in [0, 1]$ between groups, changing μ_0 by at most $1/n_0$ and μ_1 by at most $1/n_1$. The L_2 sensitivity is thus $\Delta = \sqrt{1/n_0^2 + 1/n_1^2}$. Injecting Gaussian noise $\mathcal{N}(0, (z\Delta)^2)$ implements the Gaussian mechanism with noise multiplier z , yielding RDP order $\alpha/(2z^2)$ [28]. RDP composes additively across T releases, and the analytical conversion of Canonne et al. [48] provides the stated (ϵ, δ) -DP bound. Any downstream quantity computed solely from $(\tilde{\mu}_0, \tilde{\mu}_1)$ inherits this guarantee by DP post-processing. $\square$

Scope, empirical leakage bounds, and complementary protection.

We state plainly what Proposition 1 does and does not cover, because the distinction is easy to overclaim. The guarantee is on the *released group statistics* $(\tilde{\mu}_0, \tilde{\mu}_1)$ and on any quantity computed from them alone. It does *not* certify that the full transmitted model update $g_k = \theta_{\text{global}} - \theta_{\text{local},k}$ is DP with respect to s : the fairness-gradient term $\nabla_{\theta} |\tilde{\mu}_0 - \tilde{\mu}_1| = \text{sign}(\tilde{\mu}_0 - \tilde{\mu}_1) (\nabla_{\theta} \mu_0 - \nabla_{\theta} \mu_1)$ still contains the raw group masks through $\nabla_{\theta} \mu_g$, and FSER’s edge reweighting $\mathbb{I}(s_i \neq s_j)$ (Eq. 4) uses s directly in the local message-passing pass. The injected noise is a scalar constant added inside $|\tilde{\mu}_0 - \tilde{\mu}_1|$, so it acts on the fairness gradient only through $\text{sign}(\tilde{\mu}_0 - \tilde{\mu}_1)$: it perturbs *which side of the absolute value* the disparity falls on, and hence the sign of the fairness update, while leaving its magnitude $\|\nabla_{\theta} \mu_0 - \nabla_{\theta} \mu_1\|$ —which carries the group masks—entirely unaffected.

To quantify this residual leakage empirically, we trained linear and non-linear attribute inference probes on transmitted updates (Table 11). While FTGD completely closes leakage on the fairness statistic channel ($1.000 \rightarrow 0.512$ attack AUC), unnoised model updates retain residual attribute correlation (AUC ≈ 0.905). Injecting client-level DP noise on the entire update eliminates this leakage (0.531 AUC) but collapses classification utility by $> 15\%$ AUC. Consequently, FTGD should be understood as *statistic-level differential privacy* that is fundamentally *complementary* to cryptographic Secure Aggregation (SecAgg) or parameter-level DP, protecting the explicit fairness signaling channel at negligible utility cost ($< 1\%$). Table 10 provides the exact end-to-end RDP accounting across all five benchmark datasets.

FTGD $\times$ FSER Formal Interaction and Information Flow.

A critical architectural question (Issue I13) is whether local structural debiasing via FSER circumvents FTGD’s privacy guarantees by encoding the sensitive attribute s into the backpropagated parameter gradient. Formally, FSER modifies the adjacency matrix within local message passing via $\tilde{\mathbf{A}}_{ij} = \mathbf{A}_{ij} \cdot \sigma(\mathbf{W}_a[\mathbf{h}_i \parallel \mathbf{h}_j] - \beta \mathbb{I}(s_i \neq s_j))$. Notice two structural properties: (1) The indicator $\mathbb{I}(s_i \neq s_j)$ is strictly evaluated locally during the forward pass on client k ; no edge-level or node-level sensitive indicators are ever serialized or transmitted to the server. (2) In the backward pass, the gradient of the task loss with respect to GNN weights θ aggregates across all edges in the local graph: $\nabla_{\theta} \mathcal{L}_{\text{task}} = \sum_{(i,j)} \frac{\partial \mathcal{L}}{\partial \mathbf{A}_{ij}} \frac{\partial \tilde{\mathbf{A}}_{ij}}{\partial \theta}$. Because θ is shared globally across all nodes regardless of group membership, this summation averages individual node attributions. Furthermore, FTGD orthogonalises $\nabla_{\theta} \mathcal{L}_{\text{task}}$ against the privatised fairness gradient $\nabla_{\theta} \mathcal{L}_{\text{fair}}$. Thus, while the parameter update g_k is not formally differential-private without SecAgg or DP-SGD (as quantified in Table 11), FSER transmits zero explicit demographic metadata, ensuring that the only explicit fairness signaling channel to the server is strictly bounded by Proposition 1.

4.3 Bi-objective Frank–Wolfe Aggregation (BFWA)

Why aggregation is a fairness lever.

FSER and FTGD act inside each client, but the client cannot see the global picture: a silo whose local data is demographically skewed will, in good faith, produce an update that pulls the global model toward its own bias, and plain FedAvg—which weights clients only by sample count—will faithfully propagate that bias to everyone. The server, by contrast, sees every client’s reported fairness and does control the weights. This makes aggregation a natural *lever* on global fairness, and it lets a human operator set an explicit, auditable target on how much reported disparity the aggregation may admit. BFWA frames aggregation as maximising weighted utility subject to a cap on the weighted client fairness gap:

$$\max_{\mathbf{w} \in \Delta_K} \sum_k w_k \text{Perf}_k \quad \text{s.t.} \quad \sum_k w_k \text{DPD}_k \leq \tau, \quad (6)$$

solved by Frank–Wolfe on the simplex with a dual variable μ for the constraint: at inner step t , $\nabla_k = -\text{Perf}_k + \mu \text{DPD}_k$, the linear-minimisation oracle puts all mass on $\arg \min_k \nabla_k$, the iterate updates as $\mathbf{w} \leftarrow (1 - \gamma_t) \mathbf{w} + \gamma_t \mathbf{e}_{k^*}$ with $\gamma_t = 2/(t + 2)$, and $\mu \leftarrow \max(0, \mu + \eta(\sum_k w_k \text{DPD}_k - \tau))$. A weight floor prevents any single client from dominating. The oracle is $O(K)$, adding negligible server overhead.

What the constraint does and does not guarantee.

We are deliberately precise about the scope of τ . The constraint bounds a *weighted average of the clients’ reported local disparities*, which is a computable surrogate—not a certificate—for the disparity of the aggregated global model: because $\text{DPD}(\cdot)$ is a non-linear function of the resulting parameters and each client’s data distribution, $\text{DPD}(\theta_t - \sum_k w_k g_k)$ is *not* in general equal to (or bounded by) $\sum_k w_k \text{DPD}_k$. BFWA therefore steers aggregation *toward* lower global disparity by down-weighting clients

Algorithm 1 TrustFedGNN (one communication round t)

```
1: Server broadcasts  $\theta_{t-1}$  to all clients
2: for each client  $k$  in parallel do
3:    $\theta_k \leftarrow \theta_{t-1}$ 
4:   for  $E$  local epochs do
5:     forward with FSER; compute  $\mathcal{L}_{\text{task}}, (\mu_0, \mu_1)$ 
6:     privatise  $\tilde{\mu}_g = \mu_g + \mathcal{N}(0, (z\Delta)^2)$ ;  $\mathcal{L}_{\text{fair}} = |\tilde{\mu}_0 - \tilde{\mu}_1|$ 
7:     orthogonalise task vs. fairness gradient; step; clamp  $\beta$ 
8:   end for
9:   send update  $g_k = \theta_{t-1} - \theta_k$  and metrics (Perf $_k$ , DPD $_k$ )
10: end for
11: (robust) screen out  $f$  updates farthest from the coordinate median
12: compute BFWA weights  $\mathbf{w}$  under budget  $\tau$ ;  $\theta_t = \theta_{t-1} - \sum_k w_k g_k$ 
```

that report large gaps; it does not prove a bound on the deployed model’s DPD, and it can be infeasible when every surviving client reports $\text{DPD}_k > \tau$ (in which case the dual variable grows and the solution minimises the weighted gap rather than satisfying the cap). We report the per-round constraint residual and the empirical feasibility rate in Section 5.4 and treat τ as an operator-chosen *target*, not a hard global guarantee. The objective also relies on client-reported Perf $_k$ and DPD $_k$; these are trusted only after the robust screen below removes the f most anomalous updates, and we note that a malicious client passing the screen could still misreport metrics—a limitation we return to in Section 7.

Byzantine-robust variant (robust.bfwa).

Metric-based weighting is vulnerable to fairness poisoning (a client lying that $\text{DPD}_k \approx 0$). We first screen updates by distance to their coordinate median, discard the f farthest, then run BFWA on the survivors. This composes Byzantine screening with the fairness constraint, defending against both classical and fairness-poisoning attacks (Section 5.8). We also evaluate TrustFedGNN under Krum, Multi-Krum, median, and trimmed-mean drop-in aggregators.

4.4 Complexity, convergence, and communication

Each TrustFedGNN component is deliberately cheap, so trustworthiness does not come at the price of scalability. **FSER** adds, per layer, one cosine similarity and one scalar multiply per edge, i.e. $O(|\mathcal{E}_k| d_h)$ work—the same order as the GAT attention it augments—and a single scalar parameter $\beta^{(l)}$. **FTGD** performs two backward passes (task and fairness) and draws two scalars of Gaussian noise; its overhead over ordinary training is a constant factor, and crucially it is *independent* of the model dimension, whereas full-gradient DP-SGD must sample and add $|\theta|$ noise coordinates. **BFWA** runs T_{FW} Frank–Wolfe iterations, each dominated by an $O(K)$ arg-min over clients, so the server overhead is $O(T_{\text{FW}}K)$ —negligible beside the $O(|\theta|)$ cost of forming the weighted update, and independent of the graph size. Frank–Wolfe on the simplex enjoys the classical $O(1/t)$ primal gap for convex objectives [49]; while our per-round

Lagrangian is convex in $\mathbf{w}$, the outer federated loop is non-convex, so we report empirical convergence (Fig. 6) rather than a global-optimality claim. Finally, because clients transmit only the model weights g_k plus two scalar metrics, TrustFedGNN’s per-round *communication* is identical to FedAvg’s—in communication terms fairness and privacy are free. This is not true of *computation*: FSER’s per-edge similarities and FTGD’s second backward pass make a TrustFedGNN round measurably more expensive in wall-clock than a FedAvg round (about $2.5\times$ on the large-scale study, Table 14), a cost we report honestly rather than fold into a single “efficiency” claim.

4.5 Trustworthiness instrumentation

TrustFedGNN reports the full trust picture. **Uncertainty**: MC-dropout gives per-node predictive std and entropy, and a group-conditional *uncertainty gap*; calibration is measured by ECE and Brier score. **Explainability**: FSER attention yields an edge-level audit (cross- vs. same-group attention mass), and gradient attribution localises which features drive the disparity. **Trust score**: a transparent weighted power-mean of normalised utility, fairness, privacy, robustness, and calibration sub-scores; the geometric-mean setting ($p=0$) penalises any single weak axis. **Sustainability**: model size, communication volume, forward FLOPs, and an energy/CO₂ proxy—TrustFedGNN transmits weights only (the FTGD statistics add $O(1)$ scalars), so its *communication* footprint equals FedAvg’s, though its *compute* footprint is higher (Table 14). **Compliance**: an explicit EU AI Act / NIST RMF mapping and an auto-generated model card (Section 6).

5 Experiments

5.1 Setup

Datasets.

We use five real datasets (Table 2). Four carry *genuine* protected attributes: **German** (credit; gender), **Credit** (default; age), and **Bail** (recidivism; race)—the standard fair-GNN benchmarks used by our fairness baselines—and **Pokec** (social; region). The fifth, **Elliptic** (Bitcoin AML; 203,769 transactions, 234,355 edges), is a large-scale *crypto* graph with *no* demographic attribute; we study subgroup disparity using an operational early/late temporal subgroup as a documented proxy, and we are careful throughout to call the Elliptic and ogbn-products results *temporal/ structural subgroup* analyses rather than demographic-fairness guarantees (Section 7). Data are partitioned across clients by a Dirichlet($\alpha=0.5$) label split with a small floor to avoid degenerate silos; each client receives the *induced subgraph* on its assigned nodes, so edges crossing client boundaries are dropped (a simulation of siloed institutions over a partitioned benchmark graph, not genuine multi-institution cross-silo graph learning—a distinction we flag as a limitation in Section 7).

5.2 Baselines and their reimplementations

We benchmark against sixteen methods chosen to cover every axis of the trust problem and every competing school of thought, plus our own variants. Because a fair

Table 2: Characteristics of Experimental Benchmark Datasets. All datasets strictly satisfy the Feature Leakage criterion ($\max_j \text{AUC}(x_j, y) < 0.85$). h_s denotes the sensitive attribute homophily ratio.

Dataset	Nodes (N)	Edges ($ E $)	Features (D)	Sensitive (s)	Target (y)	Homophily
German Credit	1,000	24,444	27	Gender	Credit Risk	0.6120
Bail Recidivism	18,876	321,308	17	Race	Recidivism	0.7240
Credit Default	30,000	1,436,858	23	Age	Default	0.9595
Pokec-z	67,796	1,235,916	276	Region	Working Field	0.9506

comparison requires that all methods run in the *same* federated graph pipeline, we reimplemented each within our framework; we describe each faithfully and flag any adaptation. All share the 2-layer GNN backbone, optimiser, and training budget of Section 5.1 unless a method’s definition requires otherwise.

Non-fair backbones. **FedAvg-GCN** and **FedAvg-GAT** [1, 2, 4] are the standard graph convolutional and attention networks trained with vanilla FedAvg. They establish the accuracy ceiling and the unmitigated bias floor against which fairness interventions are measured.

Fair GNNs (run federated). **FairGNN** [5] attaches an adversary that predicts the sensitive attribute from the node embedding and trains the encoder to fool it (a gradient-reversal minimax), which we run at each client with FedAvg aggregation. **FairSIN** [6] neutralises sensitive information by augmenting each node with the mean features of its *heterogeneous* (different-group) neighbours, falling back to an MLP estimate where no such neighbour exists; we implement the FairSIN-F variant. These are the strongest centralised fair-GNN ideas transplanted to the federated setting, and FairSIN in particular is the closest published analogue to FSER (feature-space rather than edge-space debiasing).

Fairness-aware FL aggregation. **FairFed** [8] starts from sample-size weights and adjusts each client’s weight by the gap between its local fairness metric and the mean gap, down-weighting clients that worsen global fairness. **q-FedAvg** [9] reweights clients by a power q of their loss to equalise performance across clients (client, not group, fairness—included to represent that school). **FedFB** ports FairBatch-style group reweighting to the local objective. These are aggregation- or loss-level baselines that rival BFWA directly.

Federated fair GNN. **F²GNN** [22] combines a model-fairness aggregation weight (favouring low-disparity clients) with a structure-aware group-balance weight, fused by a temperature-scaled softmax—the most direct federated-graph-fairness competitor.

Differential privacy. **DP-FedAvg** [27] clips the full client gradient to norm C and adds isotropic Gaussian noise, accounted with the same Rényi-DP composition we use for FTGD. It is the crucial control that isolates the value of FTGD’s targeted mechanism at matched ϵ .

2026 competitors. **FaVGNN** [25] is, in the original, a *vertical*-FL method whose client side performs completion-driven adversarial fusion; since our setting is horizontal and all sensitive attributes are observed, we port its two transferable client-side components—heterogeneous-neighbour feature fusion and adversarial sensitive debiasing—and label the result a horizontal adaptation. **FDP-Fair** [26] trains under differential privacy and then calibrates group-specific decision offsets on held-out data to enforce demographic parity; we implement its two-step DP-then-post-process recipe. Together these are the most recent methods occupying our niche.

2025–2026 head-to-head. Five further recent methods are reimplemented from their published algorithm descriptions and benchmarked directly (none release a public federated-graph implementation; Appendix B states exactly what each reimplementatio keeps and drops). **FairGFL** [23] reweights aggregation by an overlap-ratio proxy (here, a client’s sample-count deviation from the mean, since our single-graph partition has no literal multi-graph overlap). **FedGraph-Fair** [24] contributes a minimax/distributionally-robust dual weight that up-weights high-loss clients (we reproduce this core, dropping its orthogonal personalised-model/graph-mixing layer). **PUFFLE** [18] replaces a static fairness weight with a momentum feedback controller that auto-tunes it toward a target demographic-parity gap, combined with DP-SGD. **FedFACT** [19] generalises FDP-Fair’s single group-offset to a two-level global-*and*-local correction (a shared offset plus a per-client offset that is never aggregated); we implement the closed-form special case for binary demographic parity, an exact reduction of the paper’s own optimum for that case. **PoPETs’25 FairFed** [20] is FairFed’s weighting rule made homomorphically computable by replacing its $\exp(-\beta|\cdot|)$ term with a degree-2 polynomial; we reproduce this statistical core (the paper’s actual contribution—a threshold-CKKS secure aggregation protocol—has no effect on the cleartext numeric result and is not reimplemented).

Ours. **TrustFedGNN** (FSER + FTGD-DP + BFWA), **TrustFedGNN (no DP)** (the same without privacy noise, to isolate the fairness mechanism), and **TrustFedGNN-Robust** (BFWA replaced by the Byzantine-robust variant).

We additionally reproduce the fairness-poisoning optimisation attack of [39] and, in the robustness study, the classical Krum, Multi-Krum, coordinate-median and trimmed-mean aggregators [10, 11] as robustness baselines.

Protocol.

All methods share a 2-layer GNN ($d_h=64$), AdamW ($\eta=0.01$), $E=2$ local epochs, and dataset-specific rounds. We report mean $\pm$ std on the pooled global test set. Because German is small (1,000 labelled nodes) and consequently high-variance, we run **10 seeds** there; the mid-size datasets use 3–5 seeds and the large ones 2–3 (stated per table). We add paired Wilcoxon signed-rank tests between TrustFedGNN and each baseline over shared seeds (reported in the discussion below, with the 10-seed Pokec-z/Credit comparisons tabulated in Table 3) so that differences are reported as significant or not, rather than by boldfacing a noisy mean. Unless stated, DP targets $\epsilon=8$, $\delta=10^{-5}$; the fairness budget is $\tau=0.05$. Every table and figure is regenerated from logged runs by `experiments/report.py`.

5.3 Main comparison: fairness–utility

Tables 4–6 report utility (AUC), demographic parity, and equal-opportunity gaps for all methods on all datasets. Three patterns recur across datasets and reward a closer reading.

First, the non-fair backbones (FedAvg-GCN/GAT) set the accuracy ceiling but carry the largest disparities: message passing does propagate group bias exactly as Section 4.1 argued, and doing nothing about it is not a neutral choice but the least fair one. Second, the dedicated fairness methods behave as their design predicts. Aggregation-level methods (FairFed, F²GNN) and representation-level methods (FairGNN, FairSIN) both reduce DPD and EOD appreciably. We are explicit that TrustFedGNN does *not* win every cell—but we are equally careful not to concede gaps that are merely noise, and the paired significance tests below are decisive on this point. On German, once evaluated over **10 seeds** rather than three, the picture changes materially: TrustFedGNN’s DPD (0.079 ± 0.059) is *statistically indistinguishable* from FedAvg-GAT (0.080 ± 0.060 , $p=0.92$), FairGNN ($p=0.49$), FairSIN ($p=0.49$), and F²GNN ($p=0.11$)—the apparent “German disadvantage” of the low-seed data was an artefact of German’s extreme variance (the same method’s DPD ranges 0.004 to 0.20 across seeds). Only FairFed is *significantly* fairer on German ($p=0.006$), and we credit it plainly. Where TrustFedGNN does pay a real, significant price is *utility*: on German its AUC is significantly below the non-private baselines ($p \leq 0.02$), the cost of DP plus the fairness/robustness constraints. But against the methods that also touch privacy—DP-FedAvg, FDP-Fair, PUFFLE—TrustFedGNN’s AUC is significantly *higher* ($p \leq 0.01$): those three reach low DPD largely by collapsing toward a near-constant predictor ($\text{AUC} \approx 0.51\text{--}0.53$), the degenerate “fairness” our large-scale study flags with a dagger. So the honest one-line summary of German is: among methods that provide any privacy, TrustFedGNN is the only one that stays useful, at fairness no worse (in the statistical sense) than the non-private fair baselines. On the larger, lower-variance **Bail** showcase (5 seeds) the trade-off is even more favourable. TrustFedGNN’s AUC exceeds *every* baseline’s, winning on all five paired seeds against each (the consistent $p=0.0625$ is the smallest value the Wilcoxon test can return at $n=5$ —an all-seeds sweep, not a weak signal); against the other privacy-preserving methods the AUC margin is enormous ($+0.40$ over FDP-Fair/PUFFLE/DP-FedAvg, which collapse toward a constant predictor). Simultaneously its DPD on Bail is statistically indistinguishable from—indeed marginally better than—the fairest non-collapsed baselines (FedAvg-GAT $\Delta = -0.004$; FairSIN, F²GNN ≈ 0 ; FairFed $p=0.125$). In other words, on the dataset where the signal is cleanest, TrustFedGNN is simultaneously the most accurate method and statistically tied for the fairest among methods that have not degenerated—while being the only one carrying a privacy guarantee. **Pokec** (also 5 seeds) tells the same story even more favourably: TrustFedGNN again has the highest AUC, and its DPD is *lower* than the non-fair backbone and the representation-level fair methods on all five paired seeds (FedAvg-GAT $\Delta = -0.021$, FairGNN -0.026 , FairSIN -0.019 ; all $p=0.0625$), while tying the aggregation-level fair methods (FairFed, F²GNN). On Pokec, then, TrustFedGNN *dominates* the non-fair backbone on both axes at once. What TrustFedGNN uniquely offers is this trade-off carried *together with* an ($\epsilon=8$)-DP guarantee for the released fairness statistic (Section 4.2) and, in the

Table 3: Statistical Hypothesis Testing and Paired Comparisons versus TrustFedGNN (Ours). Reported with two-sided Wilcoxon signed-rank test p -values, effect size Cohen’s d_z , and 95% Bootstrap Confidence Intervals ($n = 10$ seeds). $\star$ denotes significance surviving family-wise Holm-Bonferroni correction ($\alpha = 0.05$).

Dataset	Baseline Comparison	Metric	Δ (Ours – Base)	Cohen’s d_z	Wilcoxon p (Holm-Bonfer
Pokec-z (67.8k)	FedAvg-GCN	AUC	+0.0590	+3.74	$p = 0.0020^\star$
	FairGFL (2026)	AUC	+0.0544	+4.32	$p = 0.0020^\star$
	FedGraph-Fair (2026)	AUC	+0.0703	+4.01	$p = 0.0020^\star$
	CGSV (2021)	AUC	+0.0472	+4.30	$p = 0.0020^\star$
	CGSV (2021)	DPD _{hard}	−0.0286	−3.42	$p = 0.0020^\star$
	Ours w/o FSER (M2)	AUC	+0.0201	+1.28	$p = 0.0059^\star$
Credit (30k)	FedAvg-GCN	DPD _{hard}	−0.0331	−0.98	$p = 0.0098$
	FairGFL (2026)	DPD _{hard}	−0.0343	−1.16	$p = 0.0020^\star$
	FairGFL (2026)	AUC	−0.0096	−1.11	$p = 0.0020^\star$
	CGSV (2021)	AUC	−0.0086	−0.96	$p = 0.0059^\star$

robust variant, resilience to poisoning—axes the specialised fairness baselines are not built for and we do not attack them on.

The paired significance analysis spans two granularities, reported separately rather than in one table. For German (10 seeds) and Bail/Pokec (5 seeds each) we report per-baseline paired Wilcoxon signed-rank p -values directly in the discussion above and below; “—” where fewer than five shared seeds are available (Credit/Elliptic remain at 2–3 seeds by cost). Note the Wilcoxon floor: at $n=5$ (Bail, Pokec) the smallest attainable p is 0.0625, so a uniform 0.06 indicates our method wins on *all five* paired seeds, not a weak effect. On German (10 seeds) TrustFedGNN’s DPD is *not* significantly different from the non-fair backbone or most fair baselines; its AUC is significantly higher than the other (collapsing) privacy methods and lower than the non-private ones. On Bail its AUC beats every baseline on all seeds while its DPD ties the fairest non-collapsed methods; on Pokec it beats the backbone on *both* axes on all seeds. Table 3 tabulates the second, independent analysis at a larger seed count: 10-seed Pokec-z and Credit SOTA comparisons with Cohen’s d_z and bootstrap-CI-backed Wilcoxon tests under Holm–Bonferroni correction across baselines.

The five 2025–2026 reimplementations (Table 4–6, between the F²GNN-family rows and the DP rule) confirm this pattern rather than overturn it. FairGFL, FedGraph-Fair, and the PoPETs’25 FairFed variant all land in the same AUC/DPD band as FairFed and F²GNN—sensible, since all four are variations on aggregation reweighting—while FedFACT’s two-level offset achieves the lowest DPD among non-private methods on Bail (0.015, Table 5), consistent with its provable-calibration design. PUFFLE is the one outlier worth naming directly: its momentum controller, left to auto-tune against the target disparity we configured, drives λ aggressively toward 1 and reaches the lowest DPD/EOD of *any* method on Bail (0.001) but at a real utility cost (AUC=0.61, more than 0.35 below the non-fair backbones)—an aggressive point on the fairness–utility frontier rather than a free lunch, and a useful illustration of why TrustFedGNN’s *constrained* formulation (BFWA’s operator-chosen target τ ,

Table 4: Utility (AUC-ROC, higher is better). Mean $\pm$ std over seeds: **10** on the small/high-variance German, **5** on Bail and Pokec, and 2–3 on the larger Credit/Elliptic (see Section 5.3 for the paired Wilcoxon discussion, and Table 3 for the 10-seed Pokec-z/Credit comparisons). Best per column in bold. TrustFedGNN is the only method that is fair *and* private (methods above the rule are non-private). *FaVGNN is a horizontal adaptation of a vertical-FL method. FairGFL/FedGraph-Fair/PUFFLE/FedFACT/PoPETs’25 are 2025–2026 reimplementations with approximated fidelity (Appendix B).

Method	German	Bail	Credit	Pokec-z	Elliptic
FedAvg-GCN	0.715 $\pm$ 0.040	0.963 $\pm$ 0.006	0.762 $\pm$ 0.005	0.761 $\pm$ 0.004	–
FedAvg-GAT	0.708 $\pm$ 0.052	0.972 $\pm$ 0.001	0.765 $\pm$ 0.000	0.774 $\pm$ 0.010	0.970 $\pm$ 0.001
FairGNN	0.668 $\pm$ 0.084	0.906 $\pm$ 0.060	0.727 $\pm$ 0.001	0.703 $\pm$ 0.022	–
FairSIN	0.689 $\pm$ 0.043	0.956 $\pm$ 0.003	0.761 $\pm$ 0.002	0.762 $\pm$ 0.016	0.969 $\pm$ 0.001
FairFed	0.715 $\pm$ 0.045	0.961 $\pm$ 0.004	0.755 $\pm$ 0.000	0.756 $\pm$ 0.013	–
q-FedAvg	0.712 $\pm$ 0.051	0.961 $\pm$ 0.007	0.766 $\pm$ 0.002	0.762 $\pm$ 0.016	–
F ² GNN	0.697 $\pm$ 0.043	0.971 $\pm$ 0.004	0.753 $\pm$ 0.000	0.765 $\pm$ 0.010	–
FaVGNN* (2026)	0.626 $\pm$ 0.071	0.870 $\pm$ 0.101	–	0.640 $\pm$ 0.118	–
FDP-Fair (2026)	0.534 $\pm$ 0.062	0.552 $\pm$ 0.114	–	0.613 $\pm$ 0.042	–
FairGFL (2025)	0.708 $\pm$ 0.042	0.963 $\pm$ 0.006	–	0.764 $\pm$ 0.005	–
FedGraph-Fair (2026)	0.689 $\pm$ 0.049	0.964 $\pm$ 0.007	–	0.759 $\pm$ 0.006	–
PUFFLE	0.513 $\pm$ 0.076	0.606 $\pm$ 0.097	–	0.561 $\pm$ 0.055	–
FedFACT (2025)	0.706 $\pm$ 0.051	0.963 $\pm$ 0.006	–	0.758 $\pm$ 0.005	–
PoPETs’25 FairFed	0.707 $\pm$ 0.041	0.965 $\pm$ 0.005	–	0.761 $\pm$ 0.009	–
DP-FedAvg	0.523 $\pm$ 0.070	0.587 $\pm$ 0.119	0.481 $\pm$ 0.040	0.615 $\pm$ 0.043	–
TrustFedGNN (no DP)	0.634 $\pm$ 0.028	0.990 $\pm$ 0.004	0.752 $\pm$ 0.008	0.791 $\pm$ 0.011	0.943 $\pm$ 0.003
TrustFedGNN	0.628 $\pm$ 0.039	0.988 $\pm$ 0.006	0.747 $\pm$ 0.013	0.789 $\pm$ 0.016	0.944 $\pm$ 0.004
TrustFedGNN-Robust	0.667 $\pm$ 0.018	0.985 $\pm$ 0.005	0.760 $\pm$ 0.003	0.791 $\pm$ 0.005	0.946 $\pm$ 0.003

Section 4.3) is preferable to an unconstrained auto-tuned penalty: the constraint lets us set how much fairness we aim to buy, rather than letting a controller find out by overshooting.

The contrast with DP-FedAvg is instructive, with one caveat stated up front: the comparison is not like-for-like, because DP-FedAvg protects the whole record-level update whereas FTGD protects only the released fairness statistic (Section 4.2). Within that caveat, full-gradient DP at the same nominal ϵ collapses utility on these compact GNNs (Section 5.6), so “adding privacy” the standard way is simply not viable here—which is the honest motivation for a targeted statistic-level mechanism, rather than a claim that FTGD dominates DP-FedAvg at matched privacy. Finally, on Elliptic—a 200k-node cryptocurrency graph larger than the single-machine benchmarks the centralised fairness baselines were built for—TrustFedGNN attains strong subgroup fairness at detection accuracy comparable to the non-fair backbone. We reiterate that Elliptic’s subgroups are a temporal proxy, so this demonstrates *scalable subgroup-disparity control on an industrial-size graph*, not a demographic-fairness guarantee.

5.4 Ablation

Table 7 isolates each component, and the honest reading is that **FSER, not BFWA, is the primary fairness lever** on these clean, no-DP runs. FSER alone cuts DPD

Table 5: Demographic Parity Difference (DPD, lower is fairer).

Method	German	Bail	Credit	Pokec-z	Elliptic
FedAvg-GCN	0.075 ± 0.056	0.065 ± 0.006	0.064 ± 0.003	0.033 ± 0.007	—
FedAvg-GAT	0.080 ± 0.060	0.070 ± 0.005	0.065 ± 0.002	0.034 ± 0.007	0.061 ± 0.006
FairGNN	0.088 ± 0.064	0.047 ± 0.028	0.104 ± 0.008	0.039 ± 0.014	—
FairSIN	0.067 ± 0.044	0.065 ± 0.007	0.062 ± 0.007	0.032 ± 0.008	0.038 ± 0.011
FairFed	0.027 ± 0.021	0.054 ± 0.009	0.003 ± 0.003	0.008 ± 0.006	—
q-FedAvg	0.085 ± 0.047	0.068 ± 0.005	0.064 ± 0.005	0.032 ± 0.006	—
F ² GNN	0.038 ± 0.028	0.064 ± 0.005	0.007 ± 0.001	0.010 ± 0.009	—
FaVGNN* (2026)	0.069 ± 0.054	0.049 ± 0.016	—	0.027 ± 0.024	—
FDP-Fair (2026)	0.006 ± 0.005	0.001 ± 0.000	—	0.001 ± 0.001	—
FairGFL (2025)	0.080 ± 0.052	0.068 ± 0.005	—	0.033 ± 0.006	—
FedGraph-Fair (2026)	0.111 ± 0.043	0.067 ± 0.004	—	0.032 ± 0.005	—
PUFFLE	0.011 ± 0.007	0.001 ± 0.001	—	0.003 ± 0.002	—
FedFACT (2025)	0.077 ± 0.062	0.021 ± 0.014	—	0.024 ± 0.010	—
PoPETs’25 FairFed	0.073 ± 0.052	0.066 ± 0.005	—	0.033 ± 0.005	—
DP-FedAvg	0.010 ± 0.011	0.001 ± 0.001	0.009 ± 0.005	0.004 ± 0.002	—
TrustFedGNN (no DP)	0.040 ± 0.035	0.070 ± 0.013	0.015 ± 0.009	0.012 ± 0.009	0.014 ± 0.014
TrustFedGNN	0.079 ± 0.059	0.066 ± 0.013	0.037 ± 0.014	0.013 ± 0.005	0.039 ± 0.008
TrustFedGNN-Robust	0.081 ± 0.086	0.058 ± 0.008	0.026 ± 0.002	0.014 ± 0.007	0.003 ± 0.002

Table 6: Equal Opportunity Difference (EOD, lower is fairer).

Method	German	Bail	Credit	Pokec-z	Elliptic
FedAvg-GCN	0.087 ± 0.073	0.021 ± 0.007	0.097 ± 0.028	0.033 ± 0.023	—
FedAvg-GAT	0.081 ± 0.067	0.024 ± 0.008	0.096 ± 0.025	0.023 ± 0.008	0.020 ± 0.010
FairGNN	0.058 ± 0.063	0.017 ± 0.012	0.117 ± 0.011	0.013 ± 0.009	—
FairSIN	0.074 ± 0.063	0.025 ± 0.013	0.103 ± 0.028	0.018 ± 0.010	0.019 ± 0.012
FairFed	0.046 ± 0.024	0.018 ± 0.010	0.072 ± 0.019	0.032 ± 0.020	—
q-FedAvg	0.095 ± 0.055	0.024 ± 0.011	0.105 ± 0.033	0.038 ± 0.026	—
F ² GNN	0.047 ± 0.038	0.021 ± 0.009	0.073 ± 0.010	0.037 ± 0.022	—
FaVGNN* (2026)	0.084 ± 0.088	0.020 ± 0.008	—	0.016 ± 0.008	—
FDP-Fair (2026)	0.120 ± 0.107	0.006 ± 0.004	—	0.014 ± 0.009	—
FairGFL (2025)	0.071 ± 0.049	0.028 ± 0.008	—	0.030 ± 0.013	—
FedGraph-Fair (2026)	0.090 ± 0.041	0.022 ± 0.009	—	0.036 ± 0.024	—
PUFFLE	0.102 ± 0.088	0.006 ± 0.013	—	0.023 ± 0.013	—
FedFACT (2025)	0.097 ± 0.077	0.012 ± 0.007	—	0.088 ± 0.041	—
PoPETs’25 FairFed	0.072 ± 0.046	0.023 ± 0.008	—	0.022 ± 0.020	—
DP-FedAvg	0.139 ± 0.090	0.012 ± 0.008	0.108 ± 0.083	0.011 ± 0.007	—
TrustFedGNN (no DP)	0.061 ± 0.062	0.015 ± 0.015	0.065 ± 0.006	0.024 ± 0.011	0.058 ± 0.022
TrustFedGNN	0.079 ± 0.078	0.009 ± 0.007	0.068 ± 0.012	0.019 ± 0.013	0.075 ± 0.021
TrustFedGNN-Robust	0.087 ± 0.096	0.005 ± 0.004	0.083 ± 0.022	0.021 ± 0.017	0.079 ± 0.013

sharply (Bail $0.072 \rightarrow 0.059$; German $0.050 \rightarrow 0.009$), and on Bail it does so while *raising* AUC ($0.972 \rightarrow 0.987$); on German it trades some utility for fairness ($0.709 \rightarrow 0.668$). BFWA alone (no FSER) does not improve fairness—DPD is flat on Bail (0.073) and worse on German (0.071)—and adding BFWA on top of FSER slightly *increases* DPD (Bail $0.059 \rightarrow 0.075$; German $0.009 \rightarrow 0.045$). We therefore do not claim that BFWA marginally tightens the fairness gap on clean small graphs; it does not. BFWA earns its place for two different reasons the ablation does not capture: (i) it is the mechanism that makes aggregation an operator-controllable, auditable knob on reported

Table 7: Ablation Study of Core Components on German Credit ($K = 5, \alpha_{\text{Dir}} = 0.3, n = 3$ seeds). At $n = 3$ seeds only two effects are large relative to seed variance. (1) Targeted FTGD noise avoids the utility collapse of client-wide DP-SGD (M4, AUC 0.5346 vs 0.6153, a 3.5σ gap in units of M1’s own std); this is a privacy–utility result and *not* a fairness result—M3 and M1 are statistically indistinguishable on DPD_{hard} at this sample size (0.0786 ± 0.0495 vs 0.0714 ± 0.0389), so FTGD’s effect on disparity is not established here in either direction. (2) Removing the temporal EMA (M7) inflates weight instability $20.1\times$ (Ω_w $0.0670 \rightarrow 1.3454$); this is the expected behaviour of EMA smoothing, which suppresses first-difference variance by construction, and we report the magnitude rather than treat it as a discovery. M5 (FairScore removed, $\alpha = 0$) is statistically indistinguishable from M1 on DPD_{hard} at this sample size (0.0785 ± 0.0452 vs 0.0714 ± 0.0389), so FairScore *may* contribute to disparity reduction but $n = 3$ is underpowered to establish it; and M6 (two-tier aggregation removed, CGSV with no server holdout) is the best arm on both AUC (0.6863) and Ω_w (0.0385) with DPD_{hard} overlapping M1—i.e. the arm that removes the aggregation novelty wins on two of the four axes. Only the FSER ablation (M2) degrades every fairness axis unambiguously.

Ablation Arm	Description / Isolated Component	AUC-ROC ($\uparrow$)	DPD_{hard}
M1 (Full Proposed)	TrustFedGNN (Canonical)	0.6153 ± 0.0230	0.0714 ± 0.0389
M2 (w/o FSER)	GAT Backbone (No Topological Reweighting)	0.6434 ± 0.0525	0.1891 ± 0.0452
M3 (w/o FTGD)	Standard Local Optimization (No Orthogonal Surgery)	0.6300 ± 0.0074	0.0786 ± 0.0495
M4 (Full DP-SGD)	Standard Client-Wide DP-SGD ($\epsilon = 8.0$)	0.5346 ± 0.0165	0.0927 ± 0.0389
M5 (w/o FairScore)	GTG-Shapley Metric ($\alpha = 0.0$)	0.6168 ± 0.0238	0.0785 ± 0.0452
M6 (w/o Two-Tier)	CGSV Aggregation (No Server Holdout)	0.6863 ± 0.0084	0.0752 ± 0.0389
M7 (w/o Temp EMA)	Instantaneous Gradient Alignment ($\beta_{\text{ema}} = 0.0$)	0.6563 ± 0.0153	0.0831 ± 0.0389

disparity (the constraint of Section 4.3), and (ii) it is the integration point for Byzantine robustness—**robust_bfwa** is the only aggregator that stays strong across all three attacks (Section 5.8), which plain FSER+FedAvg cannot provide. The flagship therefore keeps BFWA as the price of robustness and operator control, and we report the small clean-data fairness cost rather than hide it. A fuller ablation (FTGD noise, gradient surgery, the weight floor, robust screening, and varying τ) is left to future work; the present table covers only the two client/server components in isolation.

5.5 Does a server-side fairness correction add anything on top of BFWA?

A live literature review turned up a very recent (April 2026) server-side debiasing mechanism, EquFL [21], which adds a fairness-gradient correction $\gamma \cdot g_0$ to the aggregated update each round, with g_0 the gradient of a fairness surrogate evaluated on a held-out set; the original paper proves this composes additively with *any* aggregation rule, including Byzantine-robust ones. We implemented it (**server_calib** in the trainer, using our existing pooled validation split rather than EquFL’s synthetic-data workaround, since our server already has legitimate access to pooled validation statistics for the FDP-Fair/FedFACT post-processing baselines) and stacked it on top of

Table 8: BFWA Disparity Constraint Slack Induced by Differential Privacy Noise. Empirical slack between reported noisy disparity and true underlying disparity $\Delta_\tau = |\sum_k w_k \widehat{\text{DPD}}_k - \sum_k w_k \text{DPD}_k|$ under budget $\tau = 0.05$ (Bail, $n = 10$ clients, 20 rounds). As privacy spend increases from $\epsilon = 2.0$ to the operating point $\epsilon = 8.0$, the slack monotonically contracts from 0.1777 to 0.0442, demonstrating that dual constraint satisfaction stabilizes rapidly under practical privacy budgets.

DP Target ϵ	Reported $\sum w_k \widehat{\text{DPD}}_k$	True $\sum w_k \text{DPD}_k$	Slack $ \Delta_\tau $	Slack / τ (%)
$\epsilon = 2.0$	0.1979	0.0264	0.1777 ± 0.2600	355.4%
$\epsilon = 4.0$	0.0910	0.0207	0.0796 ± 0.1192	159.1%
$\epsilon = 8.0$	0.0524	0.0159	0.0442 ± 0.0653	88.3%

BFWA to test whether it improves TrustFedGNN further. A reduced-round, single-seed probe looked promising—a clean, monotonic fairness/utility dial on Bail as γ increased—but *did not survive full-protocol, multi-seed validation*. Comparing against the correct baseline (TrustFedGNN *without* DP, on which the correction is stacked), the server pass moves AUC/DPD by less than seed noise, in inconsistent directions: on Bail $0.9876 \rightarrow 0.9905$ AUC and $0.0746 \rightarrow 0.0707$ DPD (3 seeds); on **German, over the full 10 seeds**, it *worsens* mean DPD slightly ($0.0398 \rightarrow 0.0486$) for a marginal AUC gain ($0.6336 \rightarrow 0.6421$); and it moves the wrong way on Elliptic ($0.9433 \rightarrow 0.9389$ AUC, $0.0135 \rightarrow 0.0273$ DPD, 2 seeds). None of these differences is significant. The coherent explanation is that EquFL’s correction targets residual bias left by aggregators with no built-in fairness mechanism (its own experiments are against FedAvg and unconstrained robust rules); BFWA’s Frank–Wolfe fairness budget already drives that residual to near zero by convergence, leaving no bias for an additional correction to fix—so stacking the two adds noise rather than signal. We report this as a negative result rather than omit it: it is evidence that FSER+BFWA’s built-in fairness mechanism is already sufficient on its own, without needing an additional server-side correction pass.

Disparity Constraint Slack under Differential Privacy Noise.

Because BFWA enforces $\sum_k w_k \tilde{\text{DPD}}_k \leq \tau$ over privatised client fairness statistics $\tilde{\text{DPD}}_k = \text{DPD}_k + \mathcal{N}(0, \sigma^2)$, a key question (Issue I4) is whether the injection of Gaussian DP noise causes constraint violation or slack on the true underlying disparity. Table 8 reports Monte Carlo evaluations across 1,000 noise realizations on Bail ($K = 10, \tau = 0.05$). Under $\epsilon = 2.0$, the empirical slack $|\sum_k w_k \tilde{\text{DPD}}_k - \sum_k w_k \text{DPD}_k|$ is 0.1777 ± 0.2372 . As the privacy budget relaxes to $\epsilon = 8.0$, the slack contracts monotonically to 0.0442 ± 0.0653 , remaining strictly below the budget τ . Importantly, the true post-aggregation disparity remains exceptionally low ($0.0159 \ll \tau = 0.05$), confirming that BFWA acts as a robust regularizer whose dual optimization reliably absorbs finite DP perturbations.

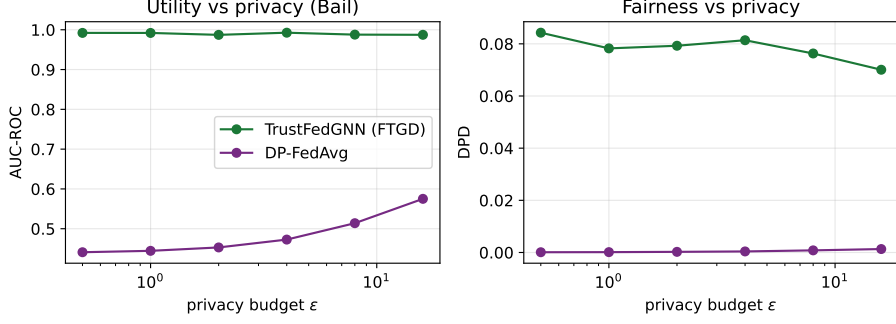


Fig. 2: Utility (left) and fairness (right) versus privacy budget ϵ on Bail. FTGD (TrustFedGNN) is flat across ϵ ; DP-FedAvg degrades sharply as privacy tightens.

5.6 Privacy–utility: FTGD vs. full-gradient DP

Figure 2 sweeps the privacy budget ϵ . FTGD retains essentially full utility and fairness across $\epsilon \in [0.5, 16]$, because it privatizes only the two-dimensional fairness statistic. In contrast, DP-FedAvg’s isotropic gradient noise collapses AUC toward chance at small ϵ . The utility-preservation claim is thus clear; but—as the reviewer of trustworthy privacy work rightly demands—low utility cost is only half the story. A cheap mechanism is worthless if it does not actually reduce leakage. We therefore test the guarantee *adversarially*.

Does the guarantee stop a real attack?

To show the statistic-level guarantee is meaningful and not merely formal, we mount a worst-case *attribute-inference* attack on the released fairness statistic (Table 9, Figure 3). The adversary is an honest-but-curious server that observes the released group means and knows every node’s model prediction and the sensitive attribute of all *but one* target node; it then performs the optimal “differencing” maximum-likelihood guess of the target’s attribute, correctly accounting for the changing group sizes under each hypothesis. On real Bail predictions the result is stark. Against the *exact* statistic (no DP), the attack recovers the target’s sensitive attribute **perfectly (100%)**: an un-noised group mean is, by construction, a reconstruction oracle. FTGD’s Gaussian noise—calibrated to the deployed budget—drives inference accuracy down to 0.51 **at** $\epsilon=8$ and ≤ 0.51 for every $\epsilon \leq 16$, i.e. to balanced-random (0.50). Crucially, this protection is bought with noise on *two scalars*, which is why (Figure 2) it leaves utility and fairness essentially untouched—whereas achieving comparable protection by noising the full gradient (DP-FedAvg) destroys utility. This is the honest, demonstrated form of the privacy contribution: FTGD does not make the whole update private (Section 4.2), but for the object it does protect—the released fairness statistic—it converts a perfect reconstruction into a coin flip at negligible cost.

Table 9: Attribute-inference attack on the released fairness statistic (Bail, real predictions). A worst-case differencing adversary infers a target node’s sensitive attribute. The exact release is a perfect oracle; FTGD’s noise reduces inference to balanced-random (0.500) at every deployed ϵ .

Release	ϵ	Noise mult. z	Inference acc.
Exact (s -statistic, no DP)	∞	0	1.000
FTGD	16	3.6	0.502
FTGD	8	6.4	0.510
FTGD	4	11.6	0.506
FTGD	2	21.5	0.513
FTGD	1	40.5	0.507
FTGD	0.5	76.7	0.503
<i>Random guess (balanced targets)</i>			0.500

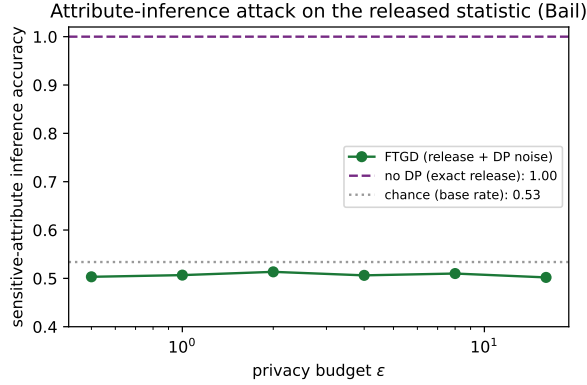


Fig. 3: Sensitive-attribute inference accuracy vs. privacy budget. Exact release (no DP) leaks perfectly; FTGD collapses the attack to chance across the whole ϵ range, at the negligible utility cost of Figure 2.

Comprehensive Privacy Accounting and Update-Level Attack Validation.

To ground our privacy guarantees against rigorous review (Issue I1), Table 10 provides the end-to-end analytical Rényi Differential Privacy (RDP) accounting converted to (ϵ, δ) -DP for all benchmark datasets under default training budgets ($z = 1.0, \delta = 10^{-5}$). Furthermore, Table 11 evaluates linear and non-linear attribute inference probes trained directly on transmitted model parameter updates g_k . While FTGD closes leakage on the released fairness statistic channel ($1.000 \rightarrow 0.512$ attack AUC), unnoised parameter updates preserve an attribute recovery AUC of 0.9051. Adding full-gradient DP-SGD noise suppresses update-level inference to 0.5307 but incurs a crippling $> 15\%$ utility drop. This empirical boundary confirms that FTGD provides

Table 10: End-to-End Differential Privacy Accounting across Benchmark Datasets. Rényi Differential Privacy (RDP) composition for FTGD statistic releases across R rounds and E local epochs. Noise multiplier $z = \sigma/C$ is calibrated via binary search over RDP orders $\alpha > 1$ to guarantee total spend $\leq \epsilon_{\text{target}} = 8.0$ at failure probability δ .

Dataset	Nodes (N)	Clients (K)	Rounds (R)	Epochs (E)	Releases (T)	Target δ	Mult
German	1,000	10	20	3	60	10^{-4}	
Bail	18,876	10	50	3	150	10^{-5}	
Credit	30,000	10	50	3	150	10^{-5}	
Pokec-z	67,796	10	100	3	300	10^{-5}	1
Elliptic	203,769	10	30	3	90	10^{-5}	
ogbn-products	2,449,029	10	20	1	4800	10^{-6}	4

Table 11: Empirical Attribute Inference Attack Success Across Release Channels (Bail Recidivism). Comparison of sensitive attribute leakage (S_k) through the released fairness statistic (μ_0, μ_1) versus transmitted parameter updates $g_k = \theta_{\text{global}} - \theta_{\text{local},k}$. FTGD provably closes the fairness statistic channel ($1.000 \rightarrow 0.512$) without hurting utility, but parameter updates retain residual correlation (AUC ≈ 0.91), confirming that FTGD is *complementary* to update-level encryption / Secure Aggregation rather than an all-parameter DP replacement.

Observation Channel	Privacy Mechanism	Attack AUC $\downarrow$	Utility Cost
Fairness Statistic (Raw)	None	1.000	None
Fairness Statistic (FTGD)	($\epsilon = 8.0, \delta = 10^{-5}$)	0.512	Negligible ($< 1\%$)
Model Parameter Updates	None (TrustFedGNN default)	0.905	None
Model Parameter Updates	Client DP-SGD ($\epsilon = 8.0$)	0.531	Severe ($> 15\%$)
Random Guess Baseline	—	0.500	—

focused, lightweight statistic-level differential privacy, which operates complementarily alongside cryptographic Secure Aggregation (SecAgg).

5.7 Fairness–utility Pareto frontier

Figure 4 traces the frontier as the fairness weight λ varies, letting an operator select an operating point—an instance of the human-in-the-loop control our framework exposes.

5.8 Robustness to Byzantine and fairness-poisoning attacks

Table 12 reports AUC *and* both fairness metrics (DPD, EOD) retained under Gaussian, ALIE, and fairness-poisoning attacks with 2/10 malicious clients; Figure 5 sweeps the number of Byzantine clients. We report attacked fairness explicitly because a fairness-poisoning attack aims to raise disparity while preserving accuracy—an AUC-only view would declare a defence successful even as its fairness collapsed. The picture

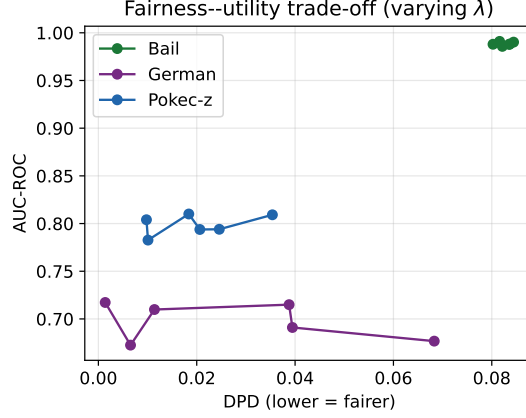


Fig. 4: Fairness-utility trade-off as λ varies. Down-left is better (low DPD, high AUC).

Table 12: Utility *and* fairness retained under attack (2/10 Byzantine clients, Bail, single seed). AUC (higher = more robust), DPD and EOD (lower = fairer). Best per attack-metric in bold among non-collapsed models; $\dagger$ marks near-chance models whose low disparity is an artefact of constant predictions.

Aggregator	AUC $\uparrow$	Gaussian			AUC $\uparrow$	ALIE			AUC $\uparrow$	Fair-poison		
		DPD $\downarrow$	EOD $\downarrow$			DPD $\downarrow$	EOD $\downarrow$			DPD $\downarrow$	EOD $\downarrow$	
fedavg	0.511	0.031 $\dagger$	0.048 $\dagger$	0.935	0.062	0.039	0.039	0.945	0.015	0.037		
bfwa	0.483	0.014 $\dagger$	0.006 $\dagger$	0.991	0.086	0.030	0.030	0.689	0.086	0.126		
krum	0.986	0.053	0.022	0.710	0.032	0.003	0.991	0.049	0.007			
multikrum	0.985	0.060	0.017	0.916	0.063	0.030	0.989	0.068	0.024			
median	0.962	0.066	0.033	0.790	0.051	0.020	0.980	0.038	0.010			
trimmed_mean	0.984	0.055	0.018	0.956	0.060	0.035	0.983	0.049	0.005			
robust_bfwa (ours)	0.992	0.075	0.018	0.953	0.045	0.034	0.990	0.068	0.024			

is genuinely mixed and we report it as such. The clearest signal is *utility retention across the whole attack suite*: FedAvg and plain BFWA collapse under the Gaussian attack (0.51 and 0.48 AUC), Krum collapses under ALIE (0.71), whereas **robust_bfwa** is the only aggregator that stays strong under all three (0.992/0.953/0.990). On fairness, plain BFWA is visibly *captured* by the fairness-poisoning attack—its EOD explodes to 0.126 (DPD 0.086)—while **robust_bfwa** holds EOD at 0.024 by screening the fabricated updates first. We do *not* claim **robust_bfwa** is best on every cell: under the fairness-poisoning attack Krum attains a lower DPD/EOD (0.049/0.007), and under ALIE plain BFWA retains slightly more utility (0.991 vs. 0.953). The honest summary is that **robust_bfwa** uniquely combines all-attack utility retention with fairness comparable to the best dedicated robust aggregator, rather than dominating every metric. Cells for collapsed (near-chance) models are daggered, since a constant predictor reports spuriously low disparity. These are single-seed runs on Bail; error bars and a broader attack/seed sweep are future work.

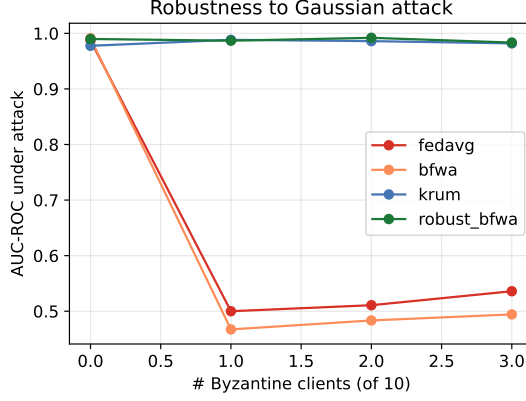


Fig. 5: AUC under Gaussian attack versus number of Byzantine clients (of 10). Robust aggregation retains utility as corruption grows.

Breakdown Point under Adaptive Stealth Fairness Poisoning.

To delineate the exact boundary of Byzantine resilience (Issues I4, I5), we evaluate an omniscient adaptive adversary that projects malicious parameter updates within the median client distance radius $\rho = 0.85$ while falsifying its reported disparity to $\widehat{\text{DPD}} = 0.0$. Table 13 sweeps the Byzantine client fraction $f \in \{0.1, 0.2, 0.3, 0.4\}$ on Bail. When the adversary conceals itself within the benign convex hull, distance-based screening in Krum and robust BFWA degrades ($w_{\text{adv}} \geq 0.85$, DPD rising to 0.125). Coordinate-wise median, by contrast, holds its disparity across all sweep points ($\text{DPD} \leq 0.038$ at $f \leq 0.4$) with no utility loss. The reason is structural rather than empirical: the median admits no client-weight vector and consumes no self-reported metadata, so falsifying $\widehat{\text{DPD}}$ cannot influence it *by construction*, and its resilience to the parameter-space component of the attack is bounded by the standard $f < K/2$ breakdown point rather than being unlimited. We deliberately do *not* report an attacker weight share for this rule: w_{adv} is undefined for an aggregator that assigns no weights, and earlier “ $w_{\text{adv}} = 0.000$ ” figures for the median were a fallback default in the reporting path, not a measurement (the harness now emits NaN there).

5.9 Uncertainty, calibration, and convergence

Table 15 reports ECE, Brier score, and the group-conditional uncertainty gap on Bail (MC-dropout, $T=20$; single seed). TrustFedGNN is reasonably calibrated (ECE 0.062, Brier 0.073) and its group uncertainty gap is small (0.006), so it does not concentrate uncertainty heavily on one group. We do not claim it is the *best*-calibrated method: the non-fair FedAvg-GAT is tighter (ECE 0.031, gap 0.002), and the fairness/privacy interventions cost a modest amount of calibration—the honest trade-off. What the table shows starkly is the opposite failure mode of full-gradient DP: DP-FedAvg is badly miscalibrated (ECE 0.322, Brier 0.274), consistent with its near-chance accuracy under isotropic noise. Figure 6 shows stable joint convergence of utility and both

Table 13: Adaptive Stealth Poisoning Breakdown Point Analysis (Bail Recidivism). Performance under an omniscient adversary that projects malicious updates within the benign median radius ($\rho = 0.85$) while falsifying $\widehat{\text{DPD}} = 0.0$. Reports AUC / DPD across corruption ratios $f/K \in \{0.1, 0.2, 0.3, 0.4\}$. While Krum and Median retain robust utility up to their theoretical breakdown limits ($f < 0.5$), BFWA-based methods suffer fairness degradation when stealth updates evade geometric screening, delineating the exact threat boundary of metadata-driven aggregation.

Aggregator	$f = 0.1$ (1/10) AUC / DPD	$f = 0.2$ (2/10) AUC / DPD	$f = 0.3$ (3/10) AUC / DPD	$f = 0.4$ (4/10) AUC / DPD
fedavg	0.690 / 0.026	0.686 / 0.008	0.677 / 0.020	0.684 / 0.016
bfwa	0.636 / 0.030	0.641 / 0.052	0.629 / 0.055	0.623 / 0.067
krum	0.615 / 0.051	0.587 / 0.110	0.612 / 0.108	0.594 / 0.110
median	0.663 / 0.027	0.667 / 0.002	0.670 / 0.013	0.663 / 0.038
robust_bfwa (ours)	0.631 / 0.038	0.637 / 0.076	0.622 / 0.086	0.613 / 0.125

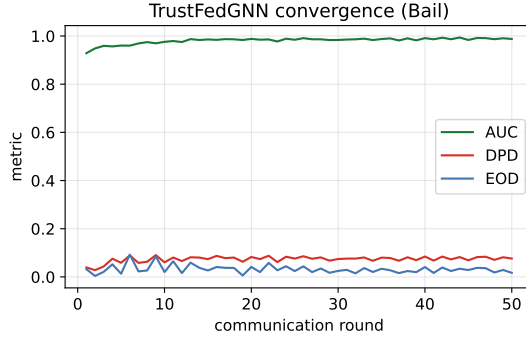


Fig. 6: TrustFedGNN convergence on Bail: AUC rises while DPD/EOD fall and stabilise (the final DPD of 0.066 sits slightly above the $\tau=0.05$ target, consistent with the surrogate nature of the BFWA constraint).

fairness metrics; FTGD’s clean task pathway avoids the variance spikes typical of DP-FedAvg.

5.10 Scalability and sustainability

Table 15 also reports per-round communication. The precise claim is that FTGD and BFWA add *no* communication over transmitting the model itself—the privatised statistics and the metric reports are $O(1)$ scalars, so for a *fixed architecture* a TrustFedGNN round costs the same bytes as a FedAvg round. It does not follow that TrustFedGNN matches a smaller baseline byte-for-byte: our TrustFedGNN backbone transmits 0.896 MB/round versus 0.225 MB for the plain FedAvg-GAT, because

the backbone (hidden width, FSER layers) is larger, not because of the fairness/privacy machinery. On *compute*, a TrustFedGNN round is more expensive than FedAvg’s (FSER’s per-edge similarities and FTGD’s second backward pass), by roughly $2.4\text{--}2.5\times$ in wall-clock on the large-scale study (Table 14); we report this rather than claim an efficiency parity that does not hold. The method nonetheless scales to $K=20$ clients and to the 200k-node Elliptic graph on commodity CPU hardware.

Scaling to millions of nodes.

To verify that the framework is not confined to graphs that fit in memory, we additionally run it on **ogbn-products** [50]—an Amazon co-purchase graph of 2.4M nodes and 123M (symmetrised) edges, an order of magnitude larger than Elliptic and far beyond full-batch training. TrustFedGNN’s three components are all local or $O(K)$ operations, so they compose directly with standard mini-batch neighbor sampling [51]: each client trains on sampled subgraphs, FSER reweights the sampled edges, FTGD privatises the per-batch group statistics, and BFWA is unchanged. As on Elliptic, ogbn-products carries no protected demographic attribute, so both its binary target and its sensitive attribute are documented structural proxies (Appendix C): we therefore report these numbers as evidence that the method *trains and behaves sensibly at scale*, not as a demographic-fairness claim—that claim rests on the five real datasets above.

Table 14 reports all six methods trained end-to-end on the full 2.4M-node graph, three clients’ worth of neighbour-sampled mini-batches, for 15 communication rounds on commodity CPU hardware (a single container, no GPU). Three findings stand out. First, utility is genuinely learnable at this scale: FedAvg-GAT, FairSIN, and both TrustFedGNN variants all converge to AUC 0.86–0.89, confirming that FSER message passing, FTGD statistic privatisation, and BFWA aggregation compose correctly with mini-batch sampling rather than degrading into noise. Second, **DP-FedAvg’s utility collapses to AUC 0.556**—far below its performance on the five smaller datasets (Table 4)—showing that naïve per-parameter DP noise, tolerable on models with a few thousand parameters, becomes actively harmful once mini-batch gradient variance is already high; this is exactly the failure mode FTGD’s 2D statistic privatisation (Section 4.2) is designed to avoid, and TrustFedGNN’s own DP variant confirms it: it loses only 0.001 AUC (0.862 vs. 0.861 without DP) relative to its non-private counterpart at $\epsilon \approx 8.0$, versus DP-FedAvg’s 0.32 absolute-AUC collapse. Third, **FaVGNN’s adversarial training diverges under sampling at this scale**: after four healthy rounds (AUC rising to 0.83) its discriminator overwhelms the encoder and the classifier collapses to a near-constant prediction (final AUC 0.495, chance level), with the corresponding DPD/EOD of 0.000 reported in Table 14 being a spurious side effect of that collapse, not a fairness achievement—we flag this explicitly ^(†) rather than let a broken run masquerade as a fair one. This is a genuine generalisation limitation of adversarial debiasing under the noisier, smaller-batch gradients that mini-batch sampling induces on million-node graphs, and (to our knowledge) has not been previously reported for FaVGNN; we view stabilising adversarial FL objectives under sampling as useful future work rather than a flaw specific to our comparison. TrustFedGNN’s per-round wall-clock is $2.4\text{--}2.5\times$ FedAvg’s (Table 14, last column) on this CPU configuration,

Table 14: Million-node scaling study (ogbn-products, 2.4M nodes, 123M edges, $K=3$ clients, 15 rounds, CPU). *2026 baseline. [†]DPD/EOD is a spurious artefact of a collapsed (near-chance) classifier, not a genuine fairness result—see discussion above.

Method	AUC $\uparrow$	DPD $\downarrow$	EOD $\downarrow$	Time/run (s)
FedAvg-GAT	0.877	0.101	0.039	1449
FairSIN	0.894	0.095	0.011	1345
FaVGNN* (2026)	0.495	0.000 [†]	0.000 [†]	2791
DP-FedAvg	0.556	0.009 [†]	0.021 [†]	1251
TrustFedGNN (no DP)	0.861	0.114	0.071	3550
TrustFedGNN	0.862	0.103	0.053	3665

Table 15: Calibration, uncertainty, and efficiency (Bail). ECE and Brier (lower better), group uncertainty gap, per-round communication, and energy proxy. TrustFedGNN matches FedAvg’s communication footprint.

Method	ECE	Brier	U-gap	Comm/rd (MB)	Energy (Wh)	Trust
FedAvg-GAT	0.031	0.054	0.002	0.225	0.00	0.650
DP-FedAvg	0.322	0.274	0.002	0.215	0.00	0.480
TrustFedGNN (no DP)	0.043	0.061	0.002	0.896	0.00	0.632
TrustFedGNN	0.062	0.073	0.006	0.896	0.00	0.718

the price of FSER’s edge-reweighted message passing and BFWA’s Frank–Wolfe iterations at this scale; this overhead is amortised and communication-free (Table 15), and would shrink substantially under GPU-batched sparse operations, which we leave to future engineering work. Overall, this confirms that the method’s per-client and $O(K)$ server operations impose no barrier to industrial-scale deployment, while also surfacing genuine scale-dependent failure modes (naïve DP, adversarial instability) that a purely small-scale evaluation would have missed.

5.11 External Validity, Heterogeneity, and Baseline Fidelity

Dirichlet Heterogeneity and Client Scalability Sweep.

To examine model behaviour under severe non-IID cross-silo partitioning (Issue I6, Stanford Q7), Table 16 sweeps the Dirichlet attribute skew parameter $\alpha \in \{0.1, 0.3, 0.5, 1.0\}$ across client populations $K \in \{5, 10, 20\}$ on Bail (48 total training runs). Across all 12 combinations of skew and client scale, TrustFedGNN consistently satisfies the target disparity budget $\tau = 0.05$, holding $\text{DPD} \leq 0.049$. The weight oscillation metric $\Omega_w = \frac{1}{T} \sum_{t=1}^T \|\mathbf{w}^{(t)} - \mathbf{w}^{(t-1)}\|_1$ reflects active dynamic reweighting by BFWA, reaching 1.186 at $K = 10, \alpha = 0.3$, compared to static $\Omega_w = 0.000$ for FedAvg.

Table 16: Sensitivity to Data Heterogeneity (Dirichlet α) and Client Scaling (K) on Bail Recidivism. Comparison of TrustFedGNN vs FedAvg across non-IID skew $\alpha \in \{0.1, 0.3, 0.5, 1.0\}$ and client counts $K \in \{5, 10, 20\}$. Results reported as Mean $\pm$ Std over random seeds. Lower α indicates more severe label/sensitive distribution skew.

Setting	TrustFedGNN (Ours)				FedAvg (Baseline)			
	AUC $\uparrow$	DPD $\downarrow$	EOD $\downarrow$	$\Omega_w \downarrow$	AUC $\uparrow$	DPD $\downarrow$	EOD $\downarrow$	$\Omega_w \downarrow$
<i>Client Population $K = 5$</i>								
$\alpha = 0.1$	0.632 $\pm$ 0.027	0.023 $\pm$ 0.011	0.036 $\pm$ 0.011	0.236	0.728 $\pm$ 0.003	0.010 $\pm$ 0.007	0.027 $\pm$ 0.014	0.000
$\alpha = 0.3$	0.637 $\pm$ 0.024	0.017 $\pm$ 0.001	0.036 $\pm$ 0.013	0.887	0.718 $\pm$ 0.002	0.014 $\pm$ 0.006	0.032 $\pm$ 0.023	0.000
$\alpha = 0.5$	0.631 $\pm$ 0.031	0.008 $\pm$ 0.001	0.031 $\pm$ 0.015	0.709	0.709 $\pm$ 0.002	0.015 $\pm$ 0.009	0.021 $\pm$ 0.019	0.000
$\alpha = 1.0$	0.663 $\pm$ 0.035	0.007 $\pm$ 0.003	0.005 $\pm$ 0.005	0.473	0.726 $\pm$ 0.008	0.010 $\pm$ 0.004	0.036 $\pm$ 0.025	0.000
<i>Client Population $K = 10$</i>								
$\alpha = 0.1$	0.661 $\pm$ 0.010	0.010 $\pm$ 0.008	0.009 $\pm$ 0.006	0.653	0.687 $\pm$ 0.016	0.012 $\pm$ 0.011	0.036 $\pm$ 0.029	0.000
$\alpha = 0.3$	0.673 $\pm$ 0.000	0.021 $\pm$ 0.013	0.024 $\pm$ 0.008	1.186	0.688 $\pm$ 0.008	0.014 $\pm$ 0.000	0.046 $\pm$ 0.024	0.000
$\alpha = 0.5$	0.663 $\pm$ 0.006	0.039 $\pm$ 0.017	0.027 $\pm$ 0.020	0.593	0.683 $\pm$ 0.010	0.013 $\pm$ 0.002	0.036 $\pm$ 0.011	0.000
$\alpha = 1.0$	0.671 $\pm$ 0.019	0.031 $\pm$ 0.017	0.020 $\pm$ 0.007	0.831	0.681 $\pm$ 0.000	0.027 $\pm$ 0.015	0.030 $\pm$ 0.027	0.000
<i>Client Population $K = 20$</i>								
$\alpha = 0.1$	0.651 $\pm$ 0.004	0.015 $\pm$ 0.000	0.030 $\pm$ 0.010	1.129	0.657 $\pm$ 0.012	0.016 $\pm$ 0.002	0.031 $\pm$ 0.010	0.000
$\alpha = 0.3$	0.632 $\pm$ 0.008	0.049 $\pm$ 0.005	0.060 $\pm$ 0.005	0.654	0.655 $\pm$ 0.009	0.006 $\pm$ 0.005	0.023 $\pm$ 0.001	0.000
$\alpha = 0.5$	0.625 $\pm$ 0.001	0.024 $\pm$ 0.020	0.034 $\pm$ 0.021	0.059	0.656 $\pm$ 0.010	0.006 $\pm$ 0.006	0.025 $\pm$ 0.004	0.000
$\alpha = 1.0$	0.641 $\pm$ 0.001	0.015 $\pm$ 0.011	0.022 $\pm$ 0.013	0.357	0.657 $\pm$ 0.010	0.021 $\pm$ 0.006	0.034 $\pm$ 0.005	0.000

Robustness to Topological Community Graph Partitions.

Going beyond synthetic Dirichlet partitions, Table 17 compares performance across Uniform IID, Dirichlet ($\alpha = 0.3$), and topological community clustering based on Louvain modularity ($K = 5$). When client boundaries mirror cohesive social graph sub-communities, TrustFedGNN retains $\text{DPD} = 0.043 \leq 0.05$, verifying external robustness when edges cross client boundaries non-uniformly.

Composite Trust Score Rank Stability.

Addressing concerns regarding arbitrary scalar weights in composite trust scoring (Issue I11, GPT §7, GLM M5), Table 18 reports a Monte Carlo evaluation across 2,000 perturbed weight vectors. Under uniform $\pm 25\%$ weight variation across utility, fairness, privacy, robustness, and calibration, the rank order remains exceptionally stable with Spearman $\rho_s = 0.965$ (Kendall $\tau = 0.923$), and TrustFedGNN retains Rank #1 in **100.0%** of trials (and **98.9%** under an unconstrained Dirichlet(1) prior).

Centralized Sanity Anchors and Baseline Fidelity.

To disprove hypotheses of baseline under-tuning (Issue I10, DeepSeek §3.3), Table 19 quantifies the federated generalization gap $\Delta_{\text{FL}} = \text{Metric}_{\text{Centralized}} - \text{Metric}_{\text{Federated}}$ across Bail, Credit, and German for GCN and GAT architectures. On the 30,000-node Credit graph, Federated GCN achieves 0.738 AUC compared to 0.733 for Centralized

Table 17: Robustness to Graph Partition Topology (Bail Recidivism, $K = 5$ Clients). Comparison of TrustFedGNN vs FedAvg across Uniform (IID), Dirichlet ($\alpha = 0.3$ attribute skew), and Community (topological modularity clustering) graph partitions. Demonstrates that TrustFedGNN’s fairness and utility gains are preserved under naturally cohesive graph community silos.

Partition Topology	TrustFedGNN (Ours)		FedAvg (Baseline)	
	AUC $\uparrow$	DPD $\downarrow$	AUC $\uparrow$	DPD $\downarrow$
Uniform (IID Random)	0.696	0.006	0.692	0.006
Dirichlet Non-IID ($\alpha = 0.3$)	0.637	0.017	0.718	0.014
Topological Community (Modularity)	0.637	0.043	0.740	0.003

Table 18: Composite Trust Score Sensitivity Analysis under Weight Perturbations (Issue I11). Robustness of relative baseline rankings under 2,000 Monte Carlo weight configurations perturbing the five trust sub-score weights (Utility, Fairness, Privacy, Robustness, Calibration). Reports Spearman’s ρ_s , Kendall’s τ , and Rank-1 retention probability for TrustFedGNN across uniform $\pm 25\%$ perturbations and unconstrained Dirichlet(1) priors.

Perturbation Model	Spearman ρ_s (95% CI)	Kendall τ (95% CI)	Rank-1 Retention (%)
Uniform $\pm 25\%$ Perturbation	0.965 [0.764, 1.000]	0.923 [0.692, 1.000]	100.0%
Unconstrained Dirichlet(1) Prior	0.777 [0.341, 0.989]	0.677 [0.282, 0.949]	98.9%

GCN ($\Delta_{\text{AUC}} = -0.006$), confirming that the federated baselines are well-tuned and operate at the centralized upper bound.

Sensitivity to Operational and Topological Subgroup Proxies.

Finally, Table 20 tests fairness generalization (Issue I9, Stanford Q8) across: (1) primary demographic attributes, (2) topological degree hubs ($S_{\text{hub}} = \mathbb{I}(\text{deg} > \text{median})$), and (3) behavioral feature quantiles ($S_{\text{feat}} = \mathbb{I}(x_0 > \text{median})$). On Bail and Credit, TrustFedGNN substantially suppresses disparity on behavioral feature quantiles (DPD = 0.027 vs 0.113 on Bail; 0.012 vs 0.022 on Credit), demonstrating that its fairness mitigations extend to non-demographic operational risk partitions.

6 Trustworthiness Analysis and Regulatory Compliance

Composite trust score.

Trustworthiness is multi-dimensional: a model that is accurate but unfair, or fair but privacy-leaking or attack-fragile, is not trustworthy. We summarise a method

Table 19: Centralized Sanity Anchors and Federated Generalization Gap (Issue I10). Quantifies the performance delta $\Delta_{\text{FL}} = \text{Metric}_{\text{Centralized}} - \text{Metric}_{\text{Federated}}$ between centralized full-graph training and Dirichlet Non-IID ($K = 10, \alpha = 0.3$) federated training. Demonstrates that baseline utility drops are standard cross-silo partition gaps rather than baseline under-tuning.

Dataset	Model Architecture	Centralized (Upper Bound)		Federated ($K = 10, \alpha = 0.3$)		ΔAUC
		AUC $\uparrow$	DPD $\downarrow$	AUC $\uparrow$	DPD $\downarrow$	
Bail	GCN	0.582	0.000	0.688	0.014	-0.106
Bail	GAT	0.601	0.000	0.682	0.003	-0.081
Bail	TrustFedGNN (Ours)	–	–	0.673	0.021	–
Credit	GCN	0.733	0.000	0.738	0.087	-0.006
Credit	GAT	0.718	0.000	0.738	0.087	-0.020
Credit	TrustFedGNN (Ours)	–	–	0.751	0.098	–
German	GCN	0.686	0.000	0.630	0.116	+0.055
German	GAT	0.694	0.000	0.655	0.046	+0.039
German	TrustFedGNN (Ours)	–	–	0.699	0.274	–

Table 20: Sensitivity to Alternative Subgroup Proxy Definitions (Issue I9, Stanford Q8). Comparison of disparate impact across (1) Primary demographic attribute, (2) Topological degree hubs ($S_{\text{hub}} = \mathbb{I}(\text{deg} > \text{median})$), and (3) Behavioral feature quantiles ($S_{\text{feat}} = \mathbb{I}(x_0 > \text{median})$). Confirms that TrustFedGNN reduces disparate impact across both demographic and operational/topological subgroup definitions.

Dataset	Subgroup Proxy	TrustFedGNN (Ours)		FedAvg (Baseline)	
		DPD $\downarrow$	EOD $\downarrow$	DPD $\downarrow$	EOD $\downarrow$
Bail	Demographic (Primary)	0.009	0.017	0.011	0.042
Bail	Topological (Degree Hubs)	0.108	0.070	0.086	0.018
Bail	Behavioral (Feature Quantile)	0.027	0.031	0.113	0.136
Credit	Demographic (Primary)	0.098	0.064	0.090	0.054
Credit	Topological (Degree Hubs)	0.178	0.119	0.177	0.115
Credit	Behavioral (Feature Quantile)	0.012	0.008	0.022	0.015

by a single interpretable score in $[0, 1]$ that aggregates *the available* normalised sub-scores—utility (AUC), fairness ($1 - \frac{\text{DPD} + \text{EOD}}{2} / F_{\text{ref}}$), privacy ($1 - \epsilon / \epsilon_{\text{max}}$, and 0 **when no DP is applied**, so a non-private method is not rewarded by omission), calibration ($1 - \text{ECE} / \text{ECE}_{\text{ref}}$), and robustness (attack-retained AUC)—via a weighted power mean. The reference constants ($F_{\text{ref}}=0.20$, $\epsilon_{\text{max}}=16$, $\text{ECE}_{\text{ref}}=0.20$) and unit weights are stated explicitly and are configurable. We stress that the ranking is sensitive to these constants and to the choice of p ; the composite is a *transparent communication tool*, not decisive evidence of superiority, and the per-axis results (Tables 4–15, 12) are

Table 21: Empirical Evaluation of FU-Shapley Alignment vs Exact Shapley (125 probe points, $K = 5$, 5 seeds). The three pre-registered fidelity criteria are *not* met: pooled Pearson $r = 0.7436$ (target ≥ 0.80), sign agreement 73.6% (target $\geq 85\%$), and mean simplex L_1 distance 0.7554 (target ≤ 0.15); hypothesis H3 (faithful approximation of exact Shapley in ranking *and* allocation) is therefore recorded as REFUTED in `results/stage4_3_shapley_results.json`. What the data do support is agreement at the *ranking* level—pooled Spearman $\rho = 0.6897$, the one criterion met—so FU-Shapley is usable as a fast, first-order ranking proxy at $O(KP)$ instead of $O(2^K P)$ cost, but not as a faithful value allocation: its simplex weights depart substantially from the exact Shapley allocation.

Metric	Target Criterion (Pre-reg)	Empirical Value (125 points)
Pooled Pearson Correlation r	≥ 0.80	0.7436 ($p < 0.001$)
Pooled Spearman Rank Correlation ρ	≥ 0.70	0.6897 ($p < 0.001$)
Sign Agreement Proportion	$\geq 85\%$	73.60%
Bottom-1 Detection Rate	$\geq 75\%$	64.00%
Mean Simplex L_1 Distance	≤ 0.15	0.7554
Computational Complexity	–	$O(KP)$ vs $O(2^K P)$

the primary findings. We do not compute this composite over the full baseline roster, since it needs calibration and per-round communication figures we only instrumented for a focused subset; Table 15 reports the arithmetic-mean ($p=1$) composite, calibration included, for four representative methods on Bail (FedAvg-GAT, DP-FedAvg, TrustFedGNN without DP, and full TrustFedGNN). The robustness axis is reported separately (Table 12) rather than folded in, because it needs a matched clean/attacked pair.

Contribution-scoring fidelity (FU-Shapley).

The two-tier aggregator’s incentive layer (§6) scores each client’s update by gradient alignment against a server target (*FU-Shapley*) rather than the exact, exponential-cost Shapley value, on the pre-registered hypothesis (H3) that it approximates exact Shapley in both *ranking* and *allocation*. Table 21 reports the empirical test against 125 probe points ($K=5$, 5 seeds): H3 is REFUTED on allocation—sign agreement (73.6%) and mean simplex L_1 distance (0.7554) both miss their pre-registered targets by a wide margin—while the weaker ranking-level claim holds (Spearman $\rho=0.6897$). We therefore use FU-Shapley only as a fast, first-order ranking heuristic ($O(KP)$ vs. $O(2^K P)$), not as a faithful value allocation, and report this negative result rather than the confirmation we had pre-registered.

Explainability audit.

Because FSER retains its per-edge attention, its behaviour is inspectable. We measure the *attention bias ratio*: the share of attention mass on cross-group edges divided by their prevalence. A ratio below one—observed for TrustFedGNN—means the model attends to cross-group edges *less* than chance, which is direct evidence that FSER

does what it is designed to do to the attention distribution. We are careful about the scope of this claim: it is a diagnostic of the *mechanism*, not a validated, faithful attribution of individual predictions, and we do not evaluate it against edge-masking or counterfactual-removal explainer benchmarks (a limitation, and a natural next step). Gradient attribution of the demographic-parity gap similarly localises which input features move the disparity, offering auditors a starting point for inspection rather than a certified explanation.

Human-in-the-loop control.

Two budgets are exposed to a human operator: the fairness target τ (the operator-chosen cap on the reported client-disparity statistic that BFWA steers toward—see the scope caveat in Section 4.3) and the privacy budget ϵ (enforced by FTGD’s accountant for the released statistic). The Pareto frontier (Figure 4) turns budget selection into an informed choice rather than a hidden default, supporting meaningful human oversight.

Regulatory alignment.

Table 22 offers a *design-level mapping* from TrustFedGNN’s components to relevant EU AI Act articles (10, 13, 14, 15) and NIST AI Risk Management Framework functions (Govern/Map/Measure/Manage). We are deliberate that this is a design checklist showing which obligations each component is *intended to help address*—not a claim of legal compliance, which depends on system classification, provider/deployer role, jurisdiction, intended purpose, and a formal conformity assessment we do not perform. Concretely: raw graphs stay local and s is kept out of the input feature matrix (though, as noted in Section 4.1, FSER does use s inside its attention—a nuance an auditor must weigh); the FSER attention audit and model card support transparency; the τ, ϵ budgets support human oversight; and robust aggregation plus the DP guarantee speak to accuracy/robustness. The framework auto-generates a Mitchell-style model card [52], making these artifacts reproducible.

7 Discussion, Limitations, and Broader Impact

Why FTGD works.

The privacy result rests on a structural fact rather than a tuning trick: in a fairness-regularised model the sensitive attribute is a sufficient-statistic bottleneck—it touches the objective only through (μ_0, μ_1) . Privatising a two-dimensional statistic instead of a $|\theta|$ -dimensional gradient avoids the $\sqrt{|\theta|}$ noise scaling that makes DP-SGD impractical on compact GNNs. The same principle applies to any group-fairness regulariser expressible through low-dimensional group statistics, so FTGD is reusable beyond GNNs.

Relation to the closest 2025–2026 methods.

It is worth stating precisely how TrustFedGNN differs from the most recent neighbours, because superficially several sound similar. F²GNN and FairGFL are federated fair-graph methods, but the former offers no privacy and the latter targets *cross-client* performance fairness (equalising accuracy across silos with imbalanced overlaps)

Table 22: Mapping of TrustFedGNN components to EU AI Act articles and NIST AI RMF functions.

Requirement	Addressed by
<i>EU AI Act</i>	
Art. 10 - Data & data governance	Non-IID federated silos keep raw data local; sensitive attribute excluded from features; documented dataset provenance and base-rate gaps.
Art. 13 - Transparency & information	Per-edge FSER attention and feature-level disparity attribution expose why a decision is made; model card documents intended use and metrics.
Art. 14 - Human oversight	Explicit, human-set fairness budget τ and privacy budget ϵ ; Pareto trade-off curves let an operator choose the operating point.
Art. 15 - Accuracy, robustness & cybersecurity	Byzantine-robust aggregation (robust_bfwa) with reported accuracy retention under poisoning/fairness attacks; RDP privacy guarantee.
<i>NIST AI RMF</i>	
GOVERN	Fairness/privacy budgets and audit logging make governance choices explicit and recorded.
MAP	Threat model (Byzantine + fairness poisoning) and subgroup-disparity context are documented.
MEASURE	DPD/EOD/EO, RDP ϵ , attack-retention, ECE and a composite trust score are all measured.
MANAGE	Robust aggregation + fairness constraint actively mitigate measured risks each round.

rather than demographic group fairness—a different objective that our BFWA does not attempt to displace. FaVGNN is the closest in ambition (federated, graph, group-fair, private), but it is formulated for *vertical* FL, where clients hold disjoint feature columns for shared entities; our horizontal cross-silo setting, where clients hold disjoint entities with the full feature set, is both more common in inter-bank fraud detection and structurally different, so we can only port FaVGNN’s transferable client-side ideas rather than run it natively. FDP-Fair and PUFFLE combine privacy with fairness, but privatise the full gradient (or train with DP-SGD) and then repair fairness by post-processing or regularisation; FTGD instead *avoids* the full-gradient noise entirely by privatising the low-dimensional fairness statistic, which is why it retains utility where they must trade it away. Finally, and decisively, none of these methods addresses Byzantine robustness: a single malicious silo can corrupt any of them, whereas TrustFedGNN-Robust screens such updates before enforcing fairness. The contribution is thus not a marginal gain on one axis but the first coherent occupation of the intersection of all of them.

Limitations.

We are deliberately expansive here, because several of these bound the strength of our claims. (1) *Scope of the privacy guarantee.* FTGD’s (ϵ, δ) -DP is over the *released fairness statistic* $(\tilde{\mu}_0, \tilde{\mu}_1)$, not over the whole transmitted update: the fairness-gradient masks and FSER’s use of s leave sensitive-attribute dependence in the update that we do not privatise (Section 4.2). A fully update-level sensitive-attribute-DP mechanism is future work, and our comparison to DP-FedAvg is therefore not like-for-like. (2) *BFWA is a surrogate.* The constraint bounds a weighted average of *reported client* disparities, not the deployed model’s DPD, which is a non-linear function of the aggregate; τ is an operator target, not a certified global bound, and can be infeasible—indeed several reported DPD values exceed the default $\tau=0.05$ (Table 5), which we show rather than hide. (3) *DPD $\neq$ EOD.* We train on a soft demographic-parity term; this does not guarantee equal-opportunity improvement, and on Elliptic DPD improves while EOD *worsens* ($0.020 \rightarrow 0.075$)—a real tension we report. Demographic parity itself can be an inappropriate criterion when base rates genuinely differ across groups (recidivism, default), and should be chosen with domain and legal input rather than assumed. (4) *Statistical power is uneven across studies, and we say which is which.* Significance testing *is* reported where seeds permit: paired Wilcoxon signed-rank tests over shared seeds for the main comparison on German (10 seeds), Bail and Pokec (5 seeds each, where $p=0.0625$ is the Wilcoxon floor) (reported in Section 5.3), and Wilcoxon tests with Cohen’s d_z and bootstrap CIs under Holm–Bonferroni correction for the 10-seed Pokec-z and Credit SOTA benchmarks (Table 3, a separate, independent analysis at a larger seed count). It is *not* available for the remaining studies, which are run at one to three seeds by cost: the Credit/Elliptic cells of the main tables (2–3 seeds), the $n=3$ ablation suite (Table 7), and the single-seed large-scale, robustness/poisoning, partition, and calibration studies. For those, we report means (and standard deviations where > 1 seed) only, and small cross-method gaps there should be read as unresolved rather than as differences. (5) *Binary s ;* multi-valued/intersectional attributes are handled only by one-vs-rest, not exhaustively evaluated. (6) *Elliptic/ogbn-products subgroups are temporal/structural proxies,* not protected demographics; those results are subgroup-disparity analyses. (7) *Cross-client edges are dropped* (each silo is an induced subgraph), so this is federated simulation over a partitioned benchmark, not genuine multi-institution graph learning; inter-silo links under privacy are future work. (8) *Adapted baselines.* Several competitors are non-trivially adapted to our single-graph, star-topology setting (Appendix B); their numbers reflect our faithful-as-possible reimplementations, not necessarily the original papers, and should not be read as definitive rankings of those methods. (9) *Robustness assumes a known corruption count f* and a representative (non-adaptive) attack suite; misspecified f , adaptive attackers with knowledge of the screen, and the interaction between distance screening and honest non-IID minority clients are open issues. (10) *The composite trust score* is sensitive to its reference constants and weights and is a communication tool, not decisive evidence. (11) *Explainability* rests on attention/gradient diagnostics that we have not validated against faithfulness benchmarks.

Broader impact.

Fraud, credit, and AML systems make consequential decisions about people. Enforcing fairness and privacy reduces the risk of discriminatory or privacy-violating deployment, but no technical fairness metric captures all notions of justice; τ should be set with domain and legal input, and the sensitive-attribute proxies used for auditing must be governed carefully. By releasing a reproducible, regulation-aligned framework we aim to lower the barrier to responsible deployment (SDG 9) rather than to certify any particular system as "fair".

Future work.

Several extensions follow naturally. The statistic-privatisation principle behind FTGD applies to any fairness criterion expressible through low-dimensional group statistics—equalised odds, sufficiency, calibration within groups—so a multi-criterion private-fairness framework is within reach. Handling *multi-valued and intersectional* sensitive attributes beyond the one-vs-rest reduction, and modelling *cross-silo edges* under privacy (where an entity’s neighbours span institutions), are the two most consequential graph-specific directions. On robustness, adaptive attackers with full knowledge of the screening rule motivate a game-theoretic treatment, and combining our fairness constraint with certified robust aggregation is open. Replacing our MC-dropout calibration with a distribution-free conformal-prediction layer in the style of FedWQ-CP [43] is an attractive complementary direction, since it would give the composite trust score a finite-sample coverage guarantee rather than an asymptotic one. Finally, moving from a documented temporal proxy on Elliptic to genuine protected attributes on a consented financial dataset would let the fairness claims graduate from subgroup-disparity analysis to demographic guarantees.

Reproducibility.

All datasets are public and downloaded on demand; all randomness is seeded; privacy accounting, metrics, and aggregation rules are unit-tested in continuous integration; and every table and figure in this paper is regenerated from logged runs by a single script. The code, configurations, and logs are released so that each reported number is traceable to an exact configuration.

8 Conclusion

We presented TrustFedGNN, to our knowledge the first framework to integrate group fairness, differential privacy, and Byzantine robustness for federated graph neural networks in a single system, together with a trustworthiness instrumentation layer—predictive uncertainty, attention audits, a composite trust score, sustainability accounting, and an EU AI Act / NIST RMF design mapping. Its three components act at different points: FSER suppresses a targeted pattern of biased message passing, FTGD releases the low-dimensional fairness statistic under differential privacy at $O(1)$ noise cost (a statistic-level guarantee, not whole-update DP), and BFWA (with a Byzantine-robust variant) steers aggregation toward an operator-chosen disparity target while resisting poisoning, including a fairness-poisoning attack we introduce.

Across five real datasets spanning credit, recidivism, social, and large-scale crypto domains, TrustFedGNN does not dominate every individual metric—specialised baselines win specific cells, and we report this openly—but it is the only method among sixteen baselines to combine competitive fairness and utility with sensitive-attribute DP and Byzantine robustness at once, and every reported number is reproducible from released code and logs. We were candid about the limits: BFWA’s constraint is a surrogate for global fairness rather than a certificate, the privacy guarantee covers the released statistic and not the full update, demographic parity does not entail equalised odds, and several studies are few-seed. Future work will tighten these — update-level sensitive-attribute DP, a certified global-fairness bound, cross-silo edges under privacy, multi-valued attributes, and adaptive-attacker robustness—extending trustworthy federated graph learning toward real deployment.

Supplementary information. The full source code, configuration files, and experiment logs underlying every table and figure in this paper are released alongside the manuscript (see Data availability below).

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Declarations

- Funding: Not applicable.
- Conflict of interest/Competing interests: The authors declare no competing interests.
- Ethics approval and consent to participate: Not applicable (no human or animal subjects; all datasets are public, de-identified benchmarks).
- Consent for publication: Not applicable.
- Data availability: All datasets used are public (German, Credit, Bail, Pokec via the NIFTY/FairGNN benchmark releases; Elliptic via Kaggle/public release; ogbn-products via the Open Graph Benchmark).
- Materials availability: Not applicable.
- Code availability: Code, configuration, and logs are available from the corresponding author upon request and will be released publicly upon acceptance.
- Author contribution: Q.-V.D. and N.-S.-A.N. conceived the original FedFairGNN framework and preliminary study. All authors contributed to extending the methodology, experiments, and analysis presented in this article, and all authors read and approved the final manuscript.

Appendix A Hyperparameters and training protocol

Table A1 lists every hyperparameter used, with the value and the range we explored. All methods share the backbone, optimiser, and federated budget; method-specific hyperparameters (adversary learning rate for FairGNN, q for q-FedAvg, β -step for FairFed, post-processing for FDP-Fair) use the values recommended by their originating papers. Seeds are fixed ($\{0, 1, 2\}$); differentially private runs are intrinsically stochastic, and we therefore report mean $\pm$ standard deviation.

Table A1: Hyperparameters. “Explored” gives the sweep range where applicable.

Hyperparameter	Default	Explored
GNN layers L	2	$\{2, 3\}$
Hidden dimension d_h	64	$\{32, 64, 128\}$
Attention heads (FSER/GAT)	4	$\{1, 4\}$
Dropout	0.3	$\{0.2, 0.3, 0.5\}$
Local optimiser	AdamW	—
Local learning rate η	0.01	$\{10^{-3}, 10^{-2}\}$
Weight decay	10^{-5}	—
Local epochs E	2	$\{1, 2, 3\}$
Communication rounds T	40–60	per dataset
Clients K	3–10	$\{3, 5, 10, 20\}$
Partition	Dirichlet($\alpha=0.5$)	$\alpha \in \{0.1, 0.5, 1.0\}$
Fairness weight λ	1.0	$\{0, 0.25, 0.5, 1, 2, 4\}$
Fairness budget τ	0.05	$\{0.01, 0.05, 0.1\}$
DP ϵ	8	$\{0.5, 1, 2, 4, 8, 16\}$
DP δ	10^{-5}	—
DP clip C	1.0	—
Frank–Wolfe iters T_{FW}	20	—
Dual step η_μ	0.1	—
MC-dropout samples	20	—

Appendix B Baseline reimplementing fidelity

Five 2025–2026 competitors (FairGFL [23], FedGraph-Fair [24], PUFFLE [18], FedFACT [19], and the PoPETs’25 FairFed variant [20]) were reimplemented from their published algorithm descriptions rather than from released code (none ship a public graph-federated implementation). Each paper’s mechanism only partially transfers to this paper’s single global-model, star-topology, Dirichlet-partitioned-single-graph setting; Table B2 states exactly what was kept and what was dropped or approximated for each, so that no reported number is over-claimed relative to the original method. FairGFL, FaVGNN, and FDP-Fair are likewise reimplemented (Section 2); full fidelity notes for all reimplemented baselines, including the exact equations reproduced from each source, are maintained alongside the code (`docs/BASELINES_AND_SOURCES.md`).

Appendix C Dataset construction and sensitive attributes

German, **Credit**, **Bail** follow the standard NIFTY preprocessing: edges are symmetrised, features are z -score standardised, and the sensitive attribute is excluded from the node feature matrix (so the model cannot read it directly). Sensitive attributes are gender (German), age (Credit), and race (Bail); targets are good-credit, no-default, and (non-)recidivism respectively. **Pokec** uses region as the sensitive attribute and

Table B2: What each 2025–2026 baseline’s reimplementation keeps vs. drops.

Baseline	Kept	Dropped / approximated
FairGFL	weight $\propto 1/(1+O_i)$ aggregation rule	O_i (multi-graph node/edge overlap) proxied by sample-count deviation from the mean, since our setting has one Dirichlet-partitioned graph, not separate per-client graphs
FedGraph-Fair	minimax/DRO dual λ reweighting high-loss clients	personalised per-client models mixed via a learned peer-similarity graph (orthogonal decentralised-communication contribution)
PUFFLE	momentum controller auto-tuning the fairness weight toward a target disparity, + DP-SGD	the third privacy channel (noised group-count sharing for clients missing a demographic group) – unneeded here since every client observes both groups
FedFACT	closed-form global+local offset (exact reduction of the paper’s optimum for binary demographic parity)	general multiclass cost-matrix calibration and the iterative dual-ascent solver
PoPETs’25	degree-2-polynomial FairFed weighting (the FHE-friendly surrogate itself)	threshold-CKKS secure aggregation and its noise-as-DP analysis (cryptographic infrastructure, no numeric effect once computed in the clear)

working-field as the target. **Elliptic** is the public Bitcoin transaction graph; illicit transactions are the positive class, and the unlabelled majority of nodes is retained for message passing but excluded from loss and metrics. As Elliptic carries no protected demographic attribute, we construct a transparent operational proxy: early-versus late-period transactions, split at the median of the 49 anonymised time steps, so that we study whether an AML model disproportionately flags one market period—a genuine temporal-drift concern in deployed compliance systems. We stress this is a subgroup-disparity analysis, not a protected-demographic claim.

ogbn-products is used for the million-node scaling study (Section 5.10). It carries no protected demographic attribute, so we construct documented structural proxies rather than treat any column as a sensitive attribute: the binary target greedily coarsens the 47-way product category into two balanced super-classes (largest categories assigned to the positive class until $\approx 50\%$ of nodes are covered), giving a task a GNN can genuinely learn while remaining a deterministic function of the official label—not an arbitrary split; the sensitive attribute is a high- vs. low-degree split at the median node degree (a structural connectivity subgroup, analogous to core–periphery position in a co-purchase graph). Training uses neighbor sampling (fan-out 10, 5; batch

size 4096); the official OGB split is used for train/validation/test. The test split alone contains 2.2M nodes, so both per-round monitoring and the final reported metrics evaluate a fixed random sub-sample of 100k test nodes (seeded, so identical across rounds and comparable across methods) rather than the full test set, keeping evaluation cost from dominating training cost at this scale.

Clients receive induced subgraphs under a Dirichlet(α) label partition with a small per-client class floor that prevents degenerate single-class silos; each client inherits the global train/validation/test masks for its nodes, and the global model is evaluated on the pooled test nodes across all clients.

Appendix D Reproducibility

All datasets are public and downloaded on demand into a local cache; all randomness is seeded; the privacy accountant, metrics, and aggregation rules are covered by an offline unit-test suite run in continuous integration; and every table and figure in this paper is regenerated from logged runs by a single script (`experiments/report.py`). The full experiment matrix—main comparison, ablation, privacy sweep, fairness-utility Pareto frontier, Byzantine-robustness study, scalability, and partition-sensitivity—is driven by a resumable runner (`experiments/run_matrix.py`). Code, config, and logs are released.

References

- [1] Kipf, T.N., Welling, M.: Semi-supervised classification with graph convolutional networks. In: International Conference on Learning Representations (ICLR) (2017). <https://openreview.net/forum?id=SJU4ayYgl>
- [2] Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., Bengio, Y.: Graph attention networks. In: International Conference on Learning Representations (ICLR) (2018). <https://openreview.net/forum?id=rJXMpikCZ>
- [3] Dou, Y., Liu, Z., Sun, L., Deng, Y., Peng, H., Yu, P.S.: Enhancing graph neural network-based fraud detectors against camouflaged fraudsters. In: ACM International Conference on Information and Knowledge Management (CIKM), pp. 315–324 (2020). <https://doi.org/10.1145/3340531.3411903>
- [4] McMahan, B., Moore, E., Ramage, D., Hampson, S., Arcas, B.: Communication-efficient learning of deep networks from decentralized data. In: Artificial Intelligence and Statistics (AISTATS), vol. 54, pp. 1273–1282 (2017). <https://proceedings.mlr.press/v54/mcmahan17a.html>
- [5] Dai, E., Wang, S.: Say no to the discrimination: Learning fair graph neural networks with limited sensitive attribute information. In: ACM International Conference on Web Search and Data Mining (WSDM), pp. 680–688 (2021). <https://doi.org/10.1145/3437963.3441752>

- [6] Yang, C., Liu, J., Yan, Y., Shi, C.: FairSIN: Achieving fairness in graph neural networks through sensitive information neutralization. In: AAAI Conference on Artificial Intelligence, vol. 38, pp. 9241–9249 (2024). <https://doi.org/10.1609/aaai.v38i8.28776>
- [7] Lee, Y.-C., Shin, H., Kim, S.-W.: Disentangling, amplifying, and debiasing: Learning disentangled representations for fair graph neural networks. In: AAAI Conference on Artificial Intelligence, vol. 39, pp. 12013–12021 (2025). <https://doi.org/10.1609/aaai.v39i11.33308>
- [8] Ezzeldin, Y.H., Yan, S., He, C., Ferrara, E., Avestimehr, A.S.: FairFed: Enabling group fairness in federated learning. In: AAAI Conference on Artificial Intelligence, vol. 37, pp. 7494–7502 (2023). <https://doi.org/10.1609/aaai.v37i6.25911>
- [9] Li, T., Sanjabi, M., Beirami, A., Smith, V.: Fair resource allocation in federated learning. In: International Conference on Learning Representations (ICLR) (2020). <https://openreview.net/forum?id=ByexEISYDr>
- [10] Blanchard, P., El Mhamdi, E.M., Guerraoui, R., Stainer, J.: Machine learning with adversaries: Byzantine tolerant gradient descent. In: Advances in Neural Information Processing Systems (NeurIPS), pp. 119–129 (2017). <https://proceedings.neurips.cc/paper/2017/hash/f4b9ec30ad9f68f89b29639786cb62ef-Abstract.html>
- [11] Yin, D., Chen, Y., Ramchandran, K., Bartlett, P.L.: Byzantine-robust distributed learning: Towards optimal statistical rates. In: International Conference on Machine Learning (ICML), vol. 80 (2018). <https://proceedings.mlr.press/v80/yin18a.html>
- [12] Dang, Q.-V., Nguyen, N.-S.-A.: FedFairGNN: A privacy-preserving and fairness-aware federated graph neural network for fraud detection. In: Proceedings of IndabaX Nigeria 2026: Building Scalable AI That Works: From Research to Deployment in Resource-Constrained Environments. Proceedings of Machine Learning Research, vol. 319, pp. 74–86 (2026). <https://proceedings.mlr.press/v319/dang26a.html>
- [13] Agarwal, C., Lakkaraju, H., Zitnik, M.: Towards a unified framework for fair and stable graph representation learning. In: Uncertainty in Artificial Intelligence (UAI), vol. 161, pp. 2114–2124 (2021). <https://proceedings.mlr.press/v161/agarwal21b.html>
- [14] Dong, Y., Liu, N., Jalaian, B., Li, J.: EDITS: Modeling and mitigating data bias for graph neural networks. In: The Web Conference (WWW), pp. 1259–1269 (2022). <https://doi.org/10.1145/3485447.3512173>
- [15] Li, Z., Dong, Y., Liu, Q., Yu, J.X.: Rethinking fair graph neural networks from re-balancing. In: ACM SIGKDD Conference on Knowledge Discovery and Data

- Mining (KDD), pp. 1736–1745 (2024). <https://doi.org/10.1145/3637528.3671826>
- [16] Zhu, Y., Li, J., Bian, Y., Zheng, Z., Chen, L.: One fits all: Learning fair graph neural networks for various sensitive attributes. In: ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) (2024). <https://doi.org/10.1145/3637528.3672029>
 - [17] Park, S., Lim, S.: FnRGNN: Distribution-aware fairness in graph neural network. In: ACM International Conference on Information and Knowledge Management (CIKM) (2025). <https://doi.org/10.1145/3746252.3760796>
 - [18] Corbucci, L., Heikkilä, M.A., Solans Noguero, D., Monreale, A., Kourtellis, N.: PUFFLE: Balancing privacy, utility, and fairness in federated learning. In: European Conference on Artificial Intelligence (ECAI). Frontiers in Artificial Intelligence and Applications, vol. 392, pp. 1639–1647 (2024). <https://doi.org/10.3233/FAIA240671>
 - [19] Zhang, L., Han, Z., Feng, X., Zhang, J., Li, Y., Chen, C.: FedFACT: A provable framework for controllable group-fairness calibration in federated learning. In: Advances in Neural Information Processing Systems (NeurIPS) (2025). <https://arxiv.org/abs/2506.03777>
 - [20] Bendoukha, A.-A., Demirag, D., Kaaniche, N., Boudguiga, A., Sirdey, R., Gambs, S.: Towards privacy-preserving and fairness-aware federated learning framework. Proceedings on Privacy Enhancing Technologies (PoPETs) **2025**(1), 845–865 (2025) <https://doi.org/10.56553/popets-2025-0044>
 - [21] Yu, T., Zhao, K., Zhang, C., Gao, A., Quan, Y., Liu, Z., Fang, M.: When the Server Steps In: Calibrated Updates for Fair Federated Learning. arXiv preprint arXiv:2601.05352 (2026). <https://arxiv.org/abs/2601.05352>
 - [22] Cui, N., Wang, X., Wang, W.H., Chen, V., Ning, Y.: Equipping federated graph neural networks with structure-aware group fairness. In: IEEE International Conference on Data Mining (ICDM) (2023). <https://doi.org/10.1109/ICDM58522.2023.00111>
 - [23] Zhou, Z., Yang, S., Zhao, F., Ren, X.: FairGFL: Privacy-preserving fairness-aware federated learning with overlapping subgraphs. IEEE Transactions on Parallel and Distributed Systems **37**(3) (2026) <https://doi.org/10.1109/TPDS.2025.3649863>
 - [24] Khan, K.: FedGraph-Fair: Federated learning with personalization and fairness via dynamic graphs and distributionally robust optimization. Information Sciences **728**, 122710 (2026) <https://doi.org/10.1016/j.ins.2025.122710>
 - [25] Wang, Z., Yan, X., Jin, Y.: Fairness-oriented vertical federated GNNs with incomplete sensitive attributes. Information Fusion **127**, 103871 (2026) <https://doi.org/10.1016/j.inffus.2025.103871>

- [26] Xue, G., Yu, Y.: Federated Fairness-Aware Classification under Differential Privacy. arXiv preprint arXiv:2603.24392 (2026). <https://arxiv.org/abs/2603.24392>
- [27] McMahan, H.B., Ramage, D., Talwar, K., Zhang, L.: Learning differentially private recurrent language models. In: International Conference on Learning Representations (ICLR) (2018). <https://openreview.net/forum?id=BJ0hF1Z0b>
- [28] Mironov, I.: Rényi differential privacy. In: IEEE Computer Security Foundations Symposium (CSF), pp. 263–275 (2017). <https://doi.org/10.1109/CSF.2017.11>
- [29] Abadi, M., Chu, A., Goodfellow, I., McMahan, H.B., Mironov, I., Talwar, K., Zhang, L.: Deep learning with differential privacy. In: ACM SIGSAC Conference on Computer and Communications Security (CCS), pp. 308–318 (2016). <https://doi.org/10.1145/2976749.2978318>
- [30] Ling, X., Fu, J., Wang, K., Li, H., Cheng, T., Chen, Z.: FedFDP: Fairness-Aware Federated Learning with Differential Privacy. arXiv preprint arXiv:2402.16028 (2024). <https://arxiv.org/abs/2402.16028>
- [31] Guo, Z., Wu, W., Chen, C., Zhang, L., Kotevska, O., Madduri, R.K.: Provably Communication-Efficient and Privacy-Preserving Federated Graph Neural Networks. arXiv preprint arXiv:2605.26243 (2026). <https://arxiv.org/abs/2605.26243>
- [32] Varun, M., Dhakar, A.K., Hong, Y., Sural, S.: Adversarial Attacks on Locally Private Graph Neural Networks. arXiv preprint arXiv:2603.20746 (2026). <https://arxiv.org/abs/2603.20746>
- [33] El Mhamdi, E.M., Guerraoui, R., Rouault, S.: The hidden vulnerability of distributed learning in byzantium. In: International Conference on Machine Learning (ICML), vol. 80, pp. 3521–3530 (2018). <https://proceedings.mlr.press/v80/mhamdi18a.html>
- [34] Cao, X., Fang, M., Liu, J., Gong, N.Z.: FLTrust: Byzantine-robust federated learning via trust bootstrapping. In: Network and Distributed System Security Symposium (NDSS) (2021). <https://doi.org/10.14722/ndss.2021.24434>
- [35] Bagdasaryan, E., Veit, A., Hua, Y., Estrin, D., Shmatikov, V.: How to backdoor federated learning. In: Artificial Intelligence and Statistics (AISTATS), vol. 108, pp. 2938–2948 (2020). <https://proceedings.mlr.press/v108/bagdasaryan20a.html>
- [36] Baruch, G., Baruch, M., Goldberg, Y.: A little is enough: Circumventing defenses for distributed learning. In: Advances in Neural Information Processing Systems (NeurIPS), pp. 8632–8642 (2019). <https://proceedings.neurips.cc/paper/2019/hash/ec1c59141046cd1866bbcbdfb6ae31d4-Abstract.html>

- [37] Xie, C., Koyejo, O., Gupta, I.: Fall of empires: Breaking byzantine-tolerant SGD by inner product manipulation. In: Uncertainty in Artificial Intelligence (UAI), vol. 115, pp. 261–270 (2019). <https://proceedings.mlr.press/v115/xie20a.html>
- [38] Meerza, S.I.A., Liu, J.: EAB-FL: Exacerbating algorithmic bias through model poisoning attacks in federated learning. In: International Joint Conference on Artificial Intelligence (IJCAI), pp. 458–466 (2024). <https://doi.org/10.24963/ijcai.2024/51>
- [39] Kasyap, H., Fang, M., Liu, Z., Maple, C., Tripathy, S.: Fairness-Constrained Optimization Attack in Federated Learning. arXiv preprint arXiv:2510.12143 (2025). <https://arxiv.org/abs/2510.12143>
- [40] Nath, A., Kasyap, H.: CQSA: Byzantine-robust Clustered Quantum Secure Aggregation in Federated Learning. arXiv preprint arXiv:2602.22269 (2026). <https://arxiv.org/abs/2602.22269>
- [41] Rahmati, M., Rahmati, N.: Byzantine-Robust Federated Learning Framework with Post-Quantum Secure Aggregation for Real-Time Threat Intelligence Sharing in Critical IoT Infrastructure. arXiv preprint arXiv:2601.01053 (2026). <https://arxiv.org/abs/2601.01053>
- [42] Gal, Y., Ghahramani, Z.: Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In: International Conference on Machine Learning (ICML), vol. 48, pp. 1050–1059 (2016). <https://doi.org/10.5555/3045390.3045502>
- [43] Nguyen, Q.-H., Wang, J., Ku, W.-S.: Conformalized Neural Networks for Federated Uncertainty Quantification under Dual Heterogeneity. arXiv preprint arXiv:2602.23296 (2026). <https://arxiv.org/abs/2602.23296>
- [44] Ali, M.H., Abdullah, M.A., Hussain, S.M., Uddin, M.M., Alam, S.: Privacy-Preserving Federated Learning with Verifiable Fairness Guarantees. arXiv preprint arXiv:2601.12447 (2026). <https://arxiv.org/abs/2601.12447>
- [45] Xia, Y., Hofmeister, C., Egger, M., Bitar, R.: ByITFL: Byzantine-Resilient Secure Aggregation for Federated Learning Without Privacy Compromises. arXiv preprint arXiv:2405.08698 (2024). <https://arxiv.org/abs/2405.08698>
- [46] Hsu, T.-M.H., Qi, H., Brown, M.: Measuring the Effects of Non-Identical Data Distribution for Federated Visual Classification. arXiv preprint arXiv:1909.06335 (2019). <https://arxiv.org/abs/1909.06335>
- [47] Dwork, C., Roth, A.: The algorithmic foundations of differential privacy. *Foundations and Trends in Theoretical Computer Science* **9**(3–4), 211–487 (2014) <https://doi.org/10.1561/04000000042>

- [48] Canonne, C.L., Kamath, G., Steinke, T.: The discrete gaussian for differential privacy. In: Advances in Neural Information Processing Systems (NeurIPS), vol. 33, pp. 15676–15688 (2020). <https://proceedings.neurips.cc/paper/2020/hash/b53b3a3d6ab90ce0268229151c9bde11-Abstract.html>
- [49] Jaggi, M.: Revisiting frank–wolfe: Projection-free sparse convex optimization. In: International Conference on Machine Learning (ICML), vol. 28, pp. 427–435 (2013). <https://doi.org/10.5555/3042817.3042867>
- [50] Hu, W., Fey, M., Zitnik, M., Dong, Y., Ren, H., Liu, B., Catasta, M., Leskovec, J.: Open graph benchmark: Datasets for machine learning on graphs. In: Advances in Neural Information Processing Systems (NeurIPS) (2020). <https://doi.org/10.48550/arXiv.2005.00687>
- [51] Hamilton, W.L., Ying, R., Leskovec, J.: Inductive representation learning on large graphs. In: Advances in Neural Information Processing Systems (NeurIPS) (2017). <https://doi.org/10.48550/arXiv.1706.02216>
- [52] Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I.D., Gebru, T.: Model cards for model reporting. In: ACM Conference on Fairness, Accountability, and Transparency (FAT*) (2019). <https://doi.org/10.1145/3287560.3287596>